CHAPTER

20 Other Two-Terminal Devices

20.1 INTRODUCTION

There are a number of two-terminal devices having a single *p-n* junction like the semiconductor or Zener diode but with different modes of operation, terminal characteristics, and areas of application. A number, including the Schottky, tunnel, varactor, photodiode, and solar cell, will be introduced in this chapter. In addition, two-terminal devices of a different construction, such as the photoconductive cell, LCD (liquid-crystal display), and thermistor, will be examined.

20.2 SCHOTTKY BARRIER (HOT-CARRIER) DIODES

In recent years, there has been increasing interest in a two-terminal device referred to as a *Schottky-barrier, surface-barrier,* or *hot-carrier* diode. Its areas of application were first limited to the very high frequency range due to its quick response time (especially important at high frequencies) and a lower noise figure (a quantity of real importance in high-frequency applications). In recent years, however, it is appearing more and more in low-voltage/high-current power supplies and ac-to-dc converters. Other areas of application of the device include radar systems, Schottky TTL logic for computers, mixers and detectors in communication equipment, instrumentation, and analog-to-digital converters.

Its construction is quite different from the conventional *p-n* junction in that a metal-semiconductor junction is created such as shown in Fig. 20.1. The semiconductor is

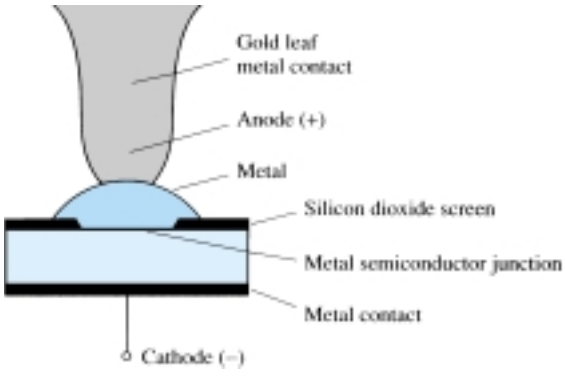


Figure 20.1 Passivated hot-carrier diode.

normally *n*-type silicon (although *p*-type silicon is sometimes used), while a host of different metals, such as molybdenum, platinum, chrome, or tungsten, are used. Different construction techniques will result in a different set of characteristics for the device, such as increased frequency range, lower forward bias, and so on. Priorities do not permit an examination of each technique here, but information will usually be provided by the manufacturer. In general, however, Schottky diode construction results in a more uniform junction region and a high level of ruggedness.

In both materials, the electron is the majority carrier. In the metal, the level of minority carriers (holes) is insignificant. When the materials are joined, the electrons in the *n*-type silicon semiconductor material immediately flow into the adjoining metal, establishing a heavy flow of majority carriers. Since the injected carriers have a very high kinetic energy level compared to the electrons of the metal, they are commonly called "hot carriers." In the conventional *p-n* junction, there was the injection of minority carriers into the adjoining region. Here the electrons are injected into a region of the same electron plurality. Schottky diodes are therefore unique in that conduction is entirely by majority carriers. The heavy flow of electrons into the metal creates a region near the junction surface depleted of carriers in the silicon material—much like the depletion region in the *p-n* junction diode. The additional carriers in the metal establish a "negative wall" in the metal at the boundary between the two materials. The net result is a "surface barrier" between the two materials, preventing any further current. That is, any electrons (negatively charged) in the silicon material face a carrier-free region and a "negative wall" at the surface of the metal.

The application of a forward bias as shown in the first quadrant of Fig. 20.2 will reduce the strength of the negative barrier through the attraction of the applied positive potential for electrons from this region. The result is a return to the heavy flow of electrons across the boundary, the magnitude of which is controlled by the level of the applied bias potential. The barrier at the junction for a Schottky diode is less than that of the *p-n* junction device in both the forward- and reverse-bias regions. The result is therefore a higher current at the same applied bias in the forward- and reverse-bias regions. This is a desirable effect in the forward-bias region but highly undesirable in the reverse-bias region.

The exponential rise in current with forward bias is described by Eq. (1.4) but with η dependent on the construction technique (1.05 for the metal whisker type of

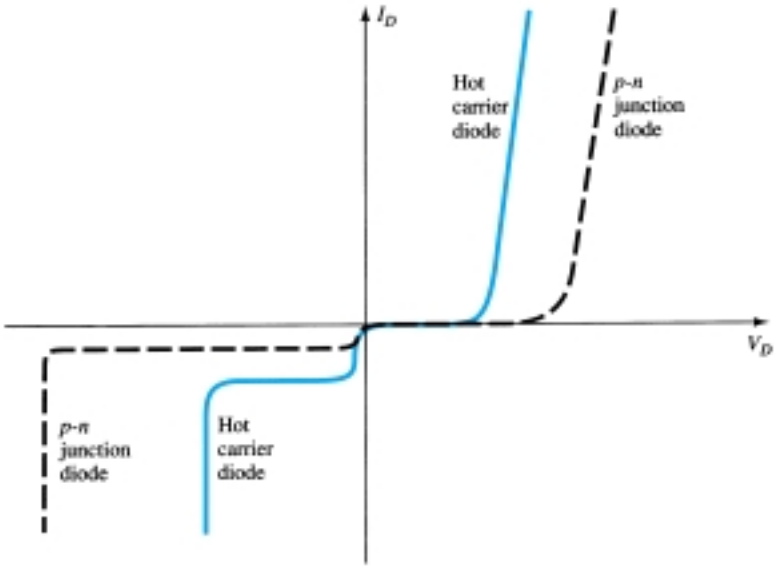


Figure 20.2 Comparison of characteristics of hot-carrier and *p-n* junction diodes.

construction, which is somewhat similar to the germanium diode). In the reverse-bias region, the current I_s is due primarily to those electrons in the metal passing into the semiconductor material. One of the areas of continuing research on the Schottky diode centers on reducing the high leakage currents that result with temperatures over 100°C. Through design, improvement units are now becoming available that have a temperature range from -65 to $+150$°C. At room temperature, I_s is typically in the microampere range for low-power units and milliampere range for high-power devices, although it is typically larger than that encountered using conventional *p-n* junction devices with the same current limits. In addition, the PIV of Schottky diodes is usually significantly less than that of a comparable *p-n* junction unit. Typically, for a 50-A unit, the PIV of the Schottky diode would be about 50 V as compared to 150 V for the *p-n* junction variety. Recent advances, however, have resulted in Schottky diodes with PIVs greater than 100 V at this current level. It is obvious from the characteristics of Fig. 20.2 that the Schottky diode is closer to the ideal set of characteristics than the point contact and has levels of V_T less than the typical silicon semiconductor *p-n* junction. The level of V_T for the "hot-carrier" diode is controlled to a large measure by the metal employed. There exists a required trade-off between temperature range and level of V_T. An increase in one appears to correspond to a resulting increase in the other. In addition, the lower the range of allowable current levels, the lower the value of V_T. For some low-level units, the value of V_T can be assumed to be essentially zero on an approximate basis. For the middle and high range, however, a value of 0.2 V would appear to be a good representative value.

The maximum current rating of the device is presently limited to about 75 A, although 100-A units appear to be on the horizon. One of the primary areas of application of this diode is in *switching power supplies* that operate in the frequency range of 20 kHz or more. A typical unit at 25°C may be rated at 50 A at a forward voltage of 0.6 V with a recovery time of 10 ns for use in one of these supplies. A *p-n* junction device with the same current limit of 50 A may have a forward voltage drop of 1.1 V and a recovery time of 30 to 50 ns. The difference in forward voltage may not appear significant, but consider the power dissipation difference: $P_{\text{hot carrier}} = (0.6\text{ V})(50\text{ A}) = 30\text{ W}$ compared to $P_{p\text{-}n} = (1.1\text{ V})(50\text{ A}) = 55\text{ W}$, which is a measurable difference when efficiency criteria must be met. There will, of course, be a higher dissipation in the reverse-bias region for the Schottky diode due to the higher leakage current, but the total power loss in the forward- and reverse-bias regions is still significantly improved as compared to the *p-n* junction device.

Recall from our discussion of reverse recovery time for the semiconductor diode that the injected minority carriers accounted for the high level of t_{rr} (the reverse recovery time). The absence of minority carriers at any appreciable level in the Schottky diode results in a reverse recovery time of significantly lower levels, as indicated above. This is the primary reason Schottky diodes are so effective at frequencies approaching 20 GHz, where the device must switch states at a very high rate. For higher frequencies the point-contact diode, with its very small junction area, is still employed.

The equivalent circuit for the device (with typical values) and a commonly used symbol appear in Fig. 20.3. A number of manufacturers prefer to use the standard diode symbol for the device since its function is essentially the same. The inductance L_P and capacitance C_P are package values, and r_B is the series resistance, which includes the contact and bulk resistance. The resistance r_d and capacitance C_J are values defined by equations introduced in earlier sections. For many applications, an excellent approximate equivalent circuit simply includes an ideal diode in parallel with the junction capacitance as shown in Fig. 20.4.

A number of hot-carrier rectifiers manufactured by Motorola Semiconductor Products, Inc., appear in Fig. 20.5 with their specifications and terminal identification.

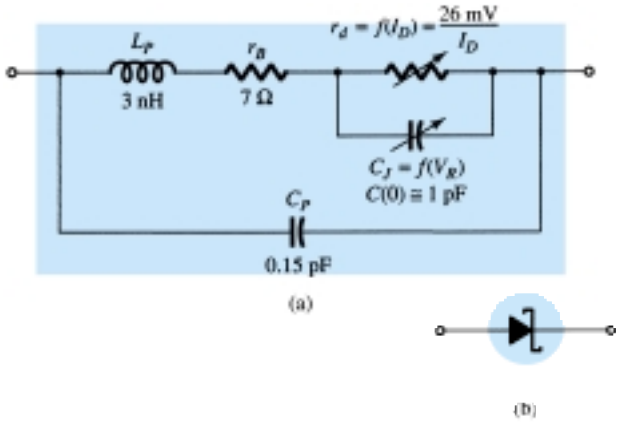


Figure 20.3 Schottky (hot-carrier) diode: (a) equivalent circuit; (b) symbol.

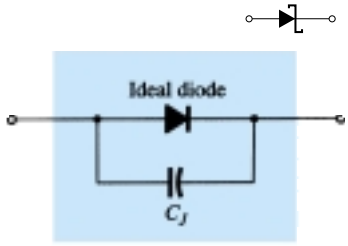
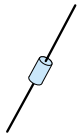


Figure 20.4 Approximate equivalent circuit for the Schottky diode.

I_o, Average rectified forward current (amperes)															
		0.5 A	1.0 A		3.0 A		3.0 A	5.0 A	15 A		25 A		40 A		
	Case	51-02 (DO-7) Glass	59-04 Plastic		267 Plastic		60 Metal		257 (DO-4) Metal				257 (DO-5) Metal		430-2 (DO-21) Metal
V_{RRM} (Volts)	Anode / Cathode:														
20		MBR020	IN5817	MBR120P	IN5820	MBR320P	MBR320M	IN5823	IN5826	MBR1520	IN5829	MBR2520	IN5832	MBR4020	MBR4020PF
30		MBR030	IN5818	MBR130P	IN5821	MBR330P	MBR330M	IN5824	IN5827	MBR1530	IN5830	MBR2530	IN5833	MBR4030	MBR4030PF
35				MBR135P		MBR335P	MBR335M			MBR1535		MBR2535		MBR4035	MBR4035PF
40			IN5819	MBR140P	IN5822	MBR340P	MBR340M	IN5825	IN5828	MBR1540	IN5831	MBR2540	IN5834	MBR4040	
I_{FSM} (Amps)		5.0	100	50	250	200	500	500	500	500	800	800	800	800	800
T_C @ Rated I_o (°C)									85	80	85	80	75	70	50
T_J Max		125°C	125°C	125°C	125°C	125°C	125°C	125°C	125°C	125°C	125°C	125°C	125°C	125°C	125°C
Max V_F @ $I_{FM} = I_o$		0.50 V	*0.60 V	0.65 V	*0.525 V	0.60 V	0.45 V@5A	*0.38 V	*0.50 V	0.55 V	*0.48 V	0.55 V	*0.59 V	0.63 V	0.63 V

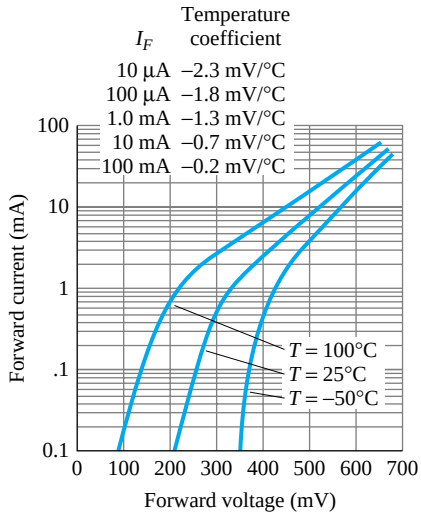
. . . Schottky barrier devices, ideal for use in low-voltage, high-frequency power supplies and as free-wheeling diodes. These units feature very low forward voltages and switching times estimated at less than 10 ns. They are offered in current ranges of 0.5 to 5.0 amperes and in voltages to 40.

V_{RRM} –respective peak reverse voltage
I_{FSM} –forward current, surge peak
I_{FM} –forward current, maximum

Figure 20.5 Motorola Schottky barrier devices. (Courtesy Motorola Semiconductor Products, Incorporated.)

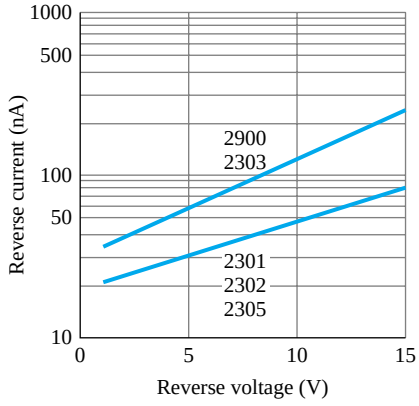
Note that the maximum forward voltage drop V_F does not exceed 0.65 V for any of the devices, while this was essentially V_T for a silicon diode.

Three sets of curves for the Hewlett-Packard 5082-2300 series of general-purpose Schottky barrier diodes are provided in Fig. 20.6. Note at $T = 100°C$ in Fig. 20.6a that V_F is only 0.1 V at a current of 0.01 mA. Note also that the reverse current has been limited to nanoamperes in Fig. 20.6b and the capacitance to 1 pF in Fig. 20.6c to ensure a high switching rate.



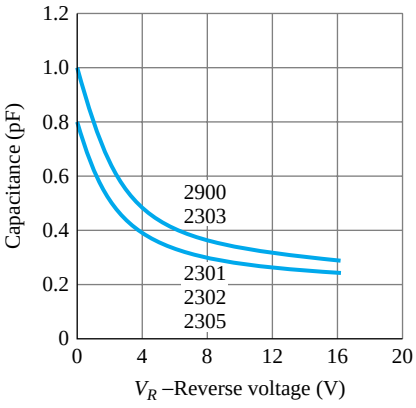
I-V Curve Showing Typical Temperature Variation for 5082-2300 Series Schottky Diodes.

(a)



5082-2300 Series Typical Reverse Current vs. Reverse Voltage at $T_A = 25°C$.

(b)



5082-2300 Series Typical Capacitance vs. Reverse Voltage at $T_A = 25°C$.

(c)

Figure 20.6 Characteristic curves for Hewlett-Packard 5082-2300 series of general-purpose Schottky barrier diodes. (Courtesy Hewlett-Packard Corporation.)

20.3 VARACTOR (VARICAP) DIODES

Varactor [also called varicap, VVC (voltage-variable capacitance), or tuning] diodes are semiconductor, voltage-dependent, variable capacitors. Their mode of operation depends on the capacitance that exists at the *p-n* junction when the element is reverse-biased. Under reverse-bias conditions, it was established that there is a region of uncovered charge on either side of the junction that together the regions make up the depletion region and define the depletion width W_d. The transition capacitance (C_T) established by the isolated uncovered charges is determined by

$$C_T = \epsilon \frac{A}{W_d} \tag{20.1}$$

where ϵ is the permittivity of the semiconductor materials, A the p-n junction area, and W_d the depletion width.

As the reverse-bias potential increases, the width of the depletion region increases, which in turn reduces the transition capacitance. The characteristics of a typical commercially available varicap diode appear in Fig. 20.7. Note the initial sharp decline in C_T with increase in reverse bias. The normal range of V_R for VVC diodes is limited to about 20 V. In terms of the applied reverse bias, the transition capacitance is given approximately by

$$C_T = \frac{K}{(V_T + V_R)^n} \tag{20.2}$$

where K = constant determined by the semiconductor material and construction technique
V_T = knee potential as defined in Section 1.6
V_R = magnitude of the applied reverse-bias potential
n = $\frac{1}{2}$ for alloy junctions and $\frac{1}{3}$ for diffused junctions

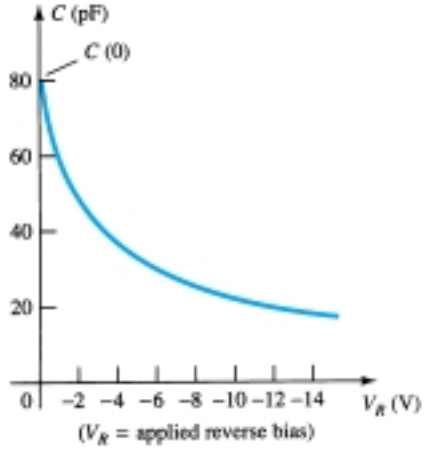


Figure 20.7 Varicap characteristics: C (pF) versus V_R.

In terms of the capacitance at the zero-bias condition $C(0)$, the capacitance as a function of V_R is given by

$$C_T(V_R) = \frac{C(0)}{(1 + |V_R/V_T|)^n} \tag{20.3}$$

The symbols most commonly used for the varicap diode and a first approximation for its equivalent circuit in the reverse-bias region are shown in Fig. 20.8. Since

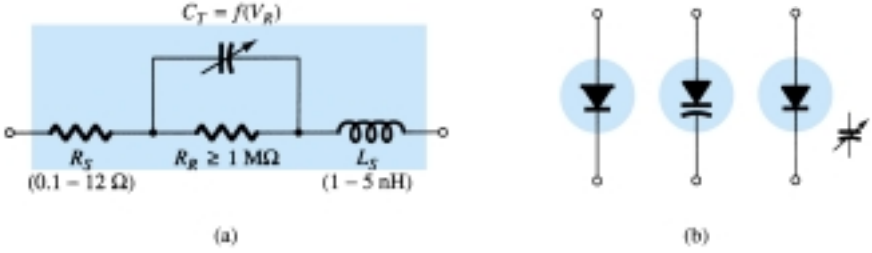


Figure 20.8 Varicap diode: (a) equivalent circuit in the reverse-bias region; (b) symbols.

we are in the reverse-bias region, the resistance in the equivalent circuit is very large in magnitude—typically 1 MΩ or larger—while R_S, the geometric resistance of the diode, is, as indicated in Fig. 20.8, very small. The magnitude of C_T will vary from about 2 to 100 pF depending on the varicap considered. To ensure that R_R is as large (for minimum leakage current) as possible, silicon is normally used in varicap diodes. The fact that the device will be employed at very high frequencies requires that we include the inductance L_S even though it is measured in nanohenries. Recall that $X_L = 2\pi fL$ and a frequency of 10 GHz with $L_S = 1$ nH will result in an $X_{LS} = 2\pi fL = (6.28)(10^{10}\text{ Hz})(10^{-9}\text{ F}) = 62.8\ \Omega$. There is obviously, therefore, a frequency limit associated with the use of each varicap diode.

Assuming the proper frequency range and a low value of R_S and X_{L_S} compared to the other series elements, then the equivalent circuit for the varicap of Fig. 20.8a can be replaced by the variable capacitor alone. The complete data sheet and its characteristic curves appear in Figs. 20.9 and 20.10, respectively. The C_3/C_{25} ratio in Fig. 20.9

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VHF/FM VARACTOR DIODE

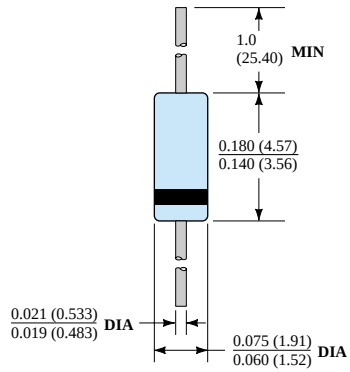
DIFFUSED SILICON PLANAR

- C_3/C_{25} . . . 5.0-6.5
- **MATCHED SETS** (Note 2)

ABSOLUTE MAXIMUM RATINGS (Note 1)

Temperatures		
Storage Temperature Range		–55°C to +150°C
Maximum Junction Operating Temperature		+150°C
Lead Temperature		+260°C
Maximum Voltage		
WIV	Working Inverse Voltage	30 V

DC-35 OUTLINE



NOTES:
Copper clad steel leads, tin plated
Gold plated leads available
Hermetically sealed glass package
Package weight is 0.14 gram

ELECTRICAL CHARACTERISTICS (25°C Ambient Temperature unless otherwise noted)

SYMBOL	CHARACTERISTIC	MIN	TYP	MAX	UNITS	TEST CONDITIONS
BV	Breakdown Voltage	30			V	$I_R = 100\ \mu A$
I_R	Reverse Current		10 0.1	50 0.5	nA μA	$V_R = 28$ V $V_R = 28$ V, $T_A = 60°C$
C	Capacitance	 4.3	29 5.1	 6.0	pF pF	$V_R = 3.0$ V, $f = 1$ MHz $V_R = 25$ V, $f = 1$ MHz
C_3/C_{25}	Capacitance Ratio	5.0	5.7	6.5		$V_R = 3$ V/25 V, $f = 1$ MHz
Q	Figure of Merit		150			$V_R = 3.0$ V, $f = 100$ MHz
R_S	Series Resistance		0.35		Ω	$C = 10$ pF, $f = 600$ MHz
L_S	Series Inductance		2.5		nH	1.5 mm from case
f_o	Series Resonant Frequency		1.4		GHz	$V_R = 25$ V

NOTES;
1. These ratings are limiting values above which the serviceability of the diodes may be impaired.
2. The capacitance diffrence between any two diodes in one set is less than 3% over the reverse voltage range of 0.5 V to 28 V

Figure 20.9 Electrical characteristics for a VHF/FM Fairchild varactor diode. (Courtesy Fairchild Camera and Instrument Corporation.)

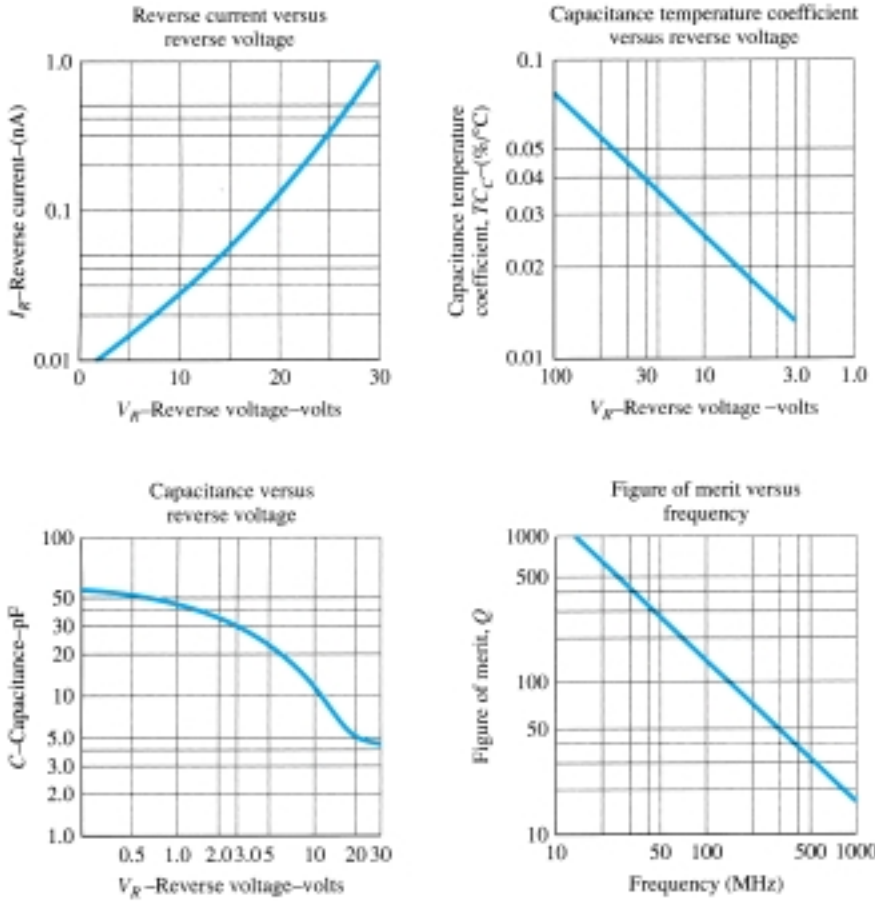


Figure 20.10 Characteristic curves for a VHF/FM Fairchild varactor diode. (Courtesy Fairchild Camera and Instrument Corporation.)

is the ratio of capacitance levels at reverse-bias potentials of 3 and 25 V. It provides a quick estimate of how much the capacitance will change with reverse-bias potential. The figure of merit is a quantity of consideration in the application of the device and is a measure of the ratio of energy stored by the capacitive device per cycle to the energy dissipated (or lost) per cycle. Since energy loss is seldom considered a positive attribute, the higher its relative value the better. The resonant frequency of the device is determined by $f_o = 1/2\pi\sqrt{LC}$ and affects the range of application of the device.

In Fig. 20.10, most quantities are self-explanatory. However, the *capacitance temperature coefficient* is defined by

$$TC_C = \frac{\Delta C}{C_0(T_1 - T_0)} \times 100\% \qquad \%/°C \qquad (20.4)$$

where ΔC is the change in capacitance due to the temperature change $T_1 - T_0$ and C_0 is the capacitance at T_0 for a particular reverse-bias potential. For example, Fig. 20.9 indicates that $C_0 = 29$ pF with $V_R = 3$ V and $T_0 = 25°C$. A change in capacitance ΔC could then be determined using Eq. (20.4) simply by substituting the new temperature T_1 and the TC_C as determined from the graph (= 0.013). At a new V_R, the value of TC_C would change accordingly. Returning to Fig. 20.9, note that the maximum frequency appearing is 600 MHz. At this frequency,

$$X_L = 2\pi fL = (6.28)(600 \times 10^6 \text{ Hz})(2.5 \times 10^{-9} \text{ F}) = 9.42\ \Omega$$

normally a quantity of sufficiently small magnitude to be ignored.

Some of the high-frequency (as defined by the small capacitance levels) areas of application include FM modulators, automatic-frequency-control devices, adjustable bandpass filters, and parametric amplifiers.

In Fig. 20.11, the varactor diode is employed in a tuning network. That is, the resonant frequency of the parallel L-C combination is determined by $f_p = 1/2\pi\sqrt{L_2 C_T'}$ (high-Q system) with the level of $C_T' = C_T + C_C$ determined by the applied reverse-bias potential V_{DD}. The coupling capacitor C_C is present to provide isolation between the shorting effect of L_2 and the applied bias. The selected frequencies of the tuned network are then passed on to the high input amplifier for further amplification.

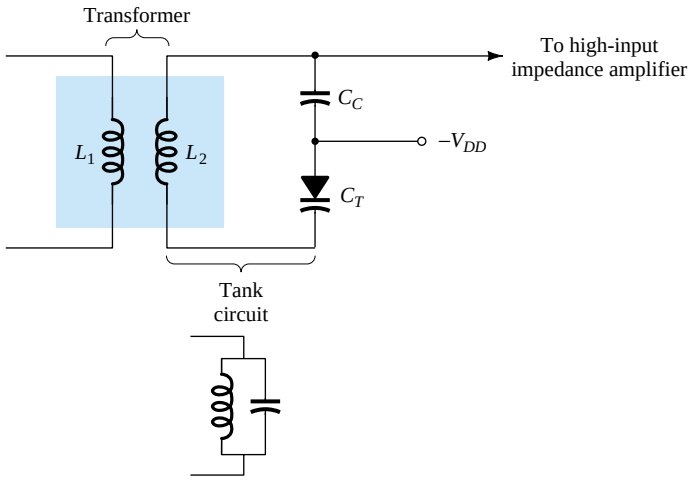


Figure 20.11 Tuning network employing a varactor diode.

20.4 POWER DIODES

There are a number of diodes designed specifically to handle the high-power and high-temperature demands of some applications. The most frequent use of power diodes occurs in the rectification process, in which ac signals (having zero average value) are converted to ones having an average or dc level. As noted in Chapter 2, when used in this capacity, diodes are normally referred to as *rectifiers.*

The majority of the power diodes are constructed using silicon because of its higher current, temperature, and PIV ratings. The higher current demands require that the junction area be larger, to ensure that there is a low forward diode resistance. If the forward resistance were too large, the I^2R losses would be excessive. The current capability of power diodes can be increased by placing two or more in parallel, and the PIV rating can be increased by stacking the diodes in series.

Various types of power diodes and their current rating have been provided in Fig. 20.12a. The high temperatures resulting from the heavy current require, in many cases, that heat sinks be used to draw the heat away from the element. A few of the various types of heat sinks available are shown in Fig. 20.12b. If heat sinks are not employed, stud diodes are designed to be attached directly to the chassis, which in turn will act as the heat sink.

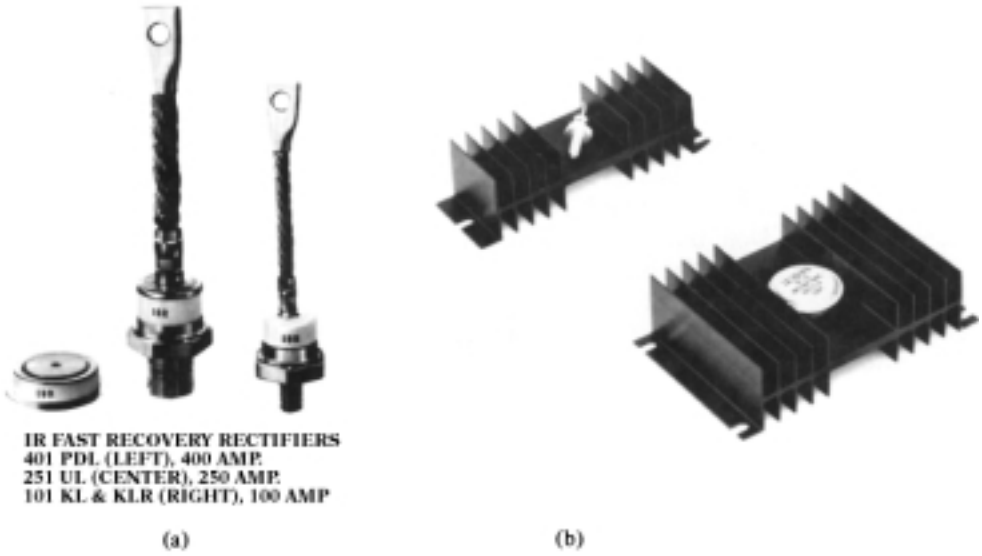


Figure 20.12 Power diodes and heat sinks. (Courtesy International Rectifier Corporation.)

20.5 TUNNEL DIODES

The tunnel diode was first introduced by Leo Esaki in 1958. Its characteristics, shown in Fig. 20.13, are different from any diode discussed thus far in that it has a negative-resistance region. In this region, an increase in terminal voltage results in a reduction in diode current.

The tunnel diode is fabricated by doping the semiconductor materials that will form the *p-n* junction at a level one hundred to several thousand times that of a typical semiconductor diode. This will result in a greatly reduced depletion region, of the order of magnitude of 10^{-6} cm, or typically about $\frac{1}{100}$ the width of this region for a typical semiconductor diode. It is this thin depletion region that many carriers can "tunnel" through, rather than attempt to surmount, at low forward-bias potentials that accounts for the peak in the curve of Fig. 20.13. For comparison purposes, a typical

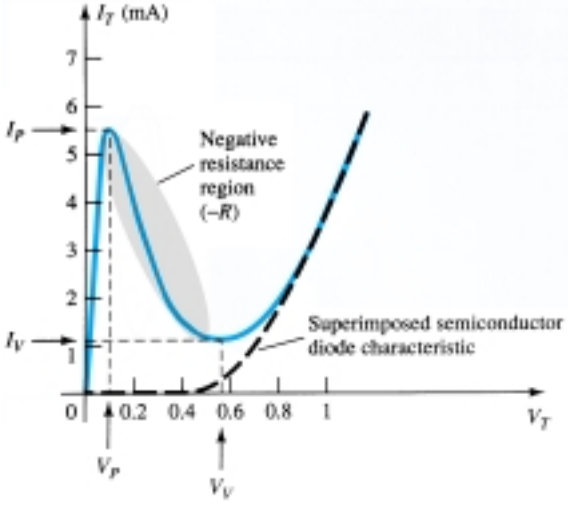


Figure 20.13 Tunnel diode characteristics.

semiconductor diode characteristic has been superimposed on the tunnel-diode characteristic of Fig. 20.13.

This reduced depletion region results in carriers "punching through" at velocities that far exceed those available with conventional diodes. The tunnel diode can therefore be used in high-speed applications such as in computers, where switching times in the order of nanoseconds or picoseconds are desirable.

You will recall from Section 1.14 that an increase in the doping level will drop the Zener potential. Note the effect of a very high doping level on this region in Fig. 20.13. The semiconductor materials most frequently used in the manufacture of tunnel diodes are germanium and gallium arsenide. The ratio I_P/I_V is very important for computer applications. For germanium, it is typically 10 : 1, while for gallium arsenide, it is closer to 20 : 1.

The peak current, I_P, of a tunnel diode can vary from a few microamperes to several hundred amperes. The peak voltage, however, is limited to about 600 mV. For this reason, a simple VOM with an internal dc battery potential of 1.5 V can severely damage a tunnel diode if applied improperly.

The tunnel diode equivalent circuit in the negative-resistance region is provided in Fig. 20.14, with the symbols most frequently employed for tunnel diodes. The values for each parameter are for the 1N2939 tunnel diode whose specifications appear in Table 20.1. The inductor L_S is due mainly to the terminal leads. The resistor R_S is due to the leads, ohmic contact at the lead–semiconductor junction, and the semiconductor materials themselves. The capacitance C is the junction diffusion capacitance, and the R is the negative resistance of the region. The negative resistance finds application in oscillators to be described later.

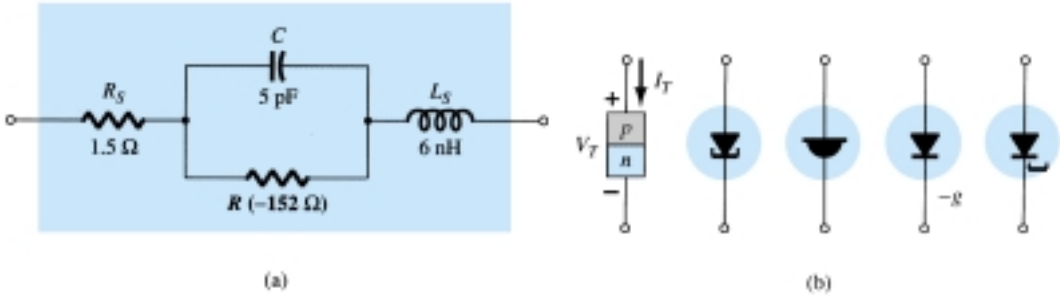


Figure 20.14 Tunnel diode: (a) equivalent circuit; (b) symbols.

TABLE 20.1 Specifications: Ge 1N2939

	Minimum	*Typical*	*Maximum*	
Absolute maximum ratings (25°C)				
Forward current (−55 to +100°C)		5 mA		
Reverse current (−55 to +100°C)		10 mA		
Electrical characteristics (25°C)($\frac{1}{8}$-in. leads)				
I_P	0.9	1.0	1.1	mA
I_V		0.1	0.14	mA
V_P	50	60	65	mV
V_V		350		mV
Reverse voltage (I_R = 1.0 mA)			30	mV
Forward peak point current voltage, V_{fp}	450	500	600	mV
I_P/I_V		10		
$-R$		−152		Ω
C		5	15	pF
L_S		6		nH
R_S		1.5	4.0	Ω

Note the lead length of $\frac{1}{8}$ in. included in the specifications. An increase in this length will cause L_S to increase. In fact, it was given for this device that L_S will vary 1 to 12 nH, depending on lead length. At high frequencies ($X_{L_S} = 2\pi f L_S$), this factor can take its toll.

The fact that V_{fp} = 500 mV (typ.) and I_{forward} (max.) = 5 mA indicates that tunnel diodes are low-power devices [P_D = (0.5 V)(5 mA) = 2.5 mW], which is also excellent for computer applications. A rendering of the device appears in Fig. 20.15.

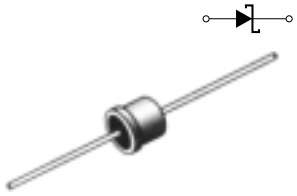


Figure 20.15 A Ge IN2939 tunnel diode. (Courtesy Powerex, Inc.)

Although the use of tunnel diodes in present-day high-frequency systems has been dramatically stalled by manufacturing techniques that suggest alternatives to the tunnel diode, its simplicity, linearity, low power drain, and reliability ensure its continued life and application. The basic construction of an advance design tunnel diode appears in Fig. 20.16 with a photograph of the actual junction.

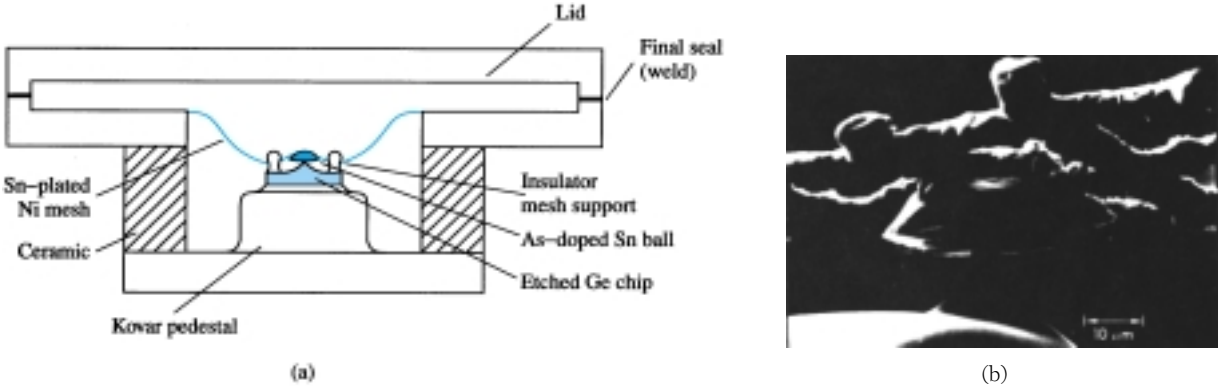


Figure 20.16 Tunnel diode: (a) construction; (b) photograph. (Courtesy *COM SAT Technical Review,* P. F. Varadi and T. D. Kirkendall.)

In Fig. 20.17, the chosen supply voltage and load resistance have defined a load line that intersects the tunnel diode characteristics at three points. Keep in mind that the load line is determined solely by the network and the characteristics by the device. The intersections at *a* and *b* are referred to as *stable* operating points, due to the positive resistance characteristic. That is, at either of these operating points, a slight disturbance in the network will not set the network into oscillations or result in a significant change in the location of the *Q*-point. For instance, if the defined operating point is at *b*, a slight increase in supply voltage *E* will move the operating point up the curve since the voltage across the diode will increase. Once the disturbance has

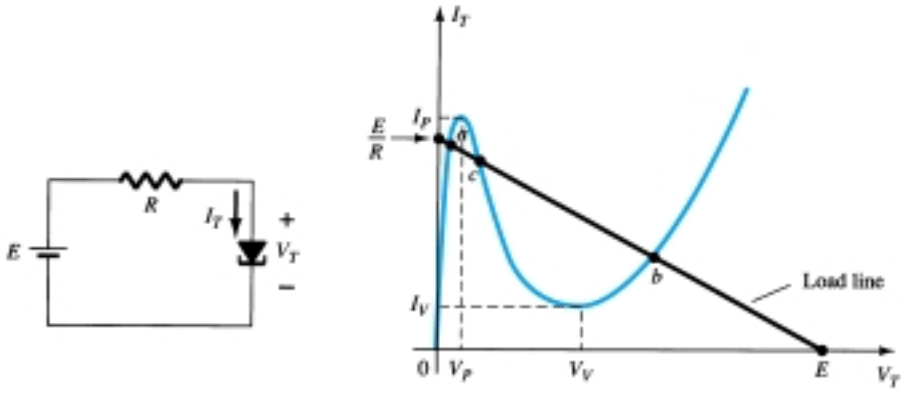


Figure 20.17 Tunnel diode and resulting load line.

passed, the voltage across the diode and the associated diode current will return to the levels defined by the *Q*-point at *b*. The operating point defined by *c* is an *unstable* one because a slight change in the voltage across or current through the diode will result in the *Q*-point moving to either *a* or *b*. For instance, the slightest increase in *E* will cause the voltage across the tunnel diode to increase above its level at *c*. In this region, however, an increase in V_T will cause a decrease in I_T and a further increase in V_T. This increased level in V_T will result in a continuing decrease in I_T, and so on. The result is an increase in V_T and a change in I_T until the stable operating point at *b* is established. A slight drop in supply voltage would result in a transition to stability at point *a*. In other words, point *c* can be defined as the operating point using the load-line technique, but once the system is energized, it will eventually stabilize at location *a* or *b*.

The availability of a negative resistance region can be put to good use in the design of oscillators, switching networks, pulse generators, and amplifiers.

In Fig. 20.18a, a *negative-resistance oscillator* was constructed using a tunnel diode. The choice of network elements is designed to establish a load line such as shown in Fig. 20.18b. Note that the only intersection with the characteristics is in the unstable negative-resistance region—a stable operating point is not defined. When the power is turned on, the terminal voltage of the supply will build up from 0 V to a final value of *E* volts. Initially, the current I_T will increase from 0 mA to I_P, resulting in a storage of energy in the inductor in the form of a magnetic field. However, once I_P is reached, the diode characteristics suggest that the current I_T must now decrease with increase in voltage across the diode. This is a contradiction to the fact that

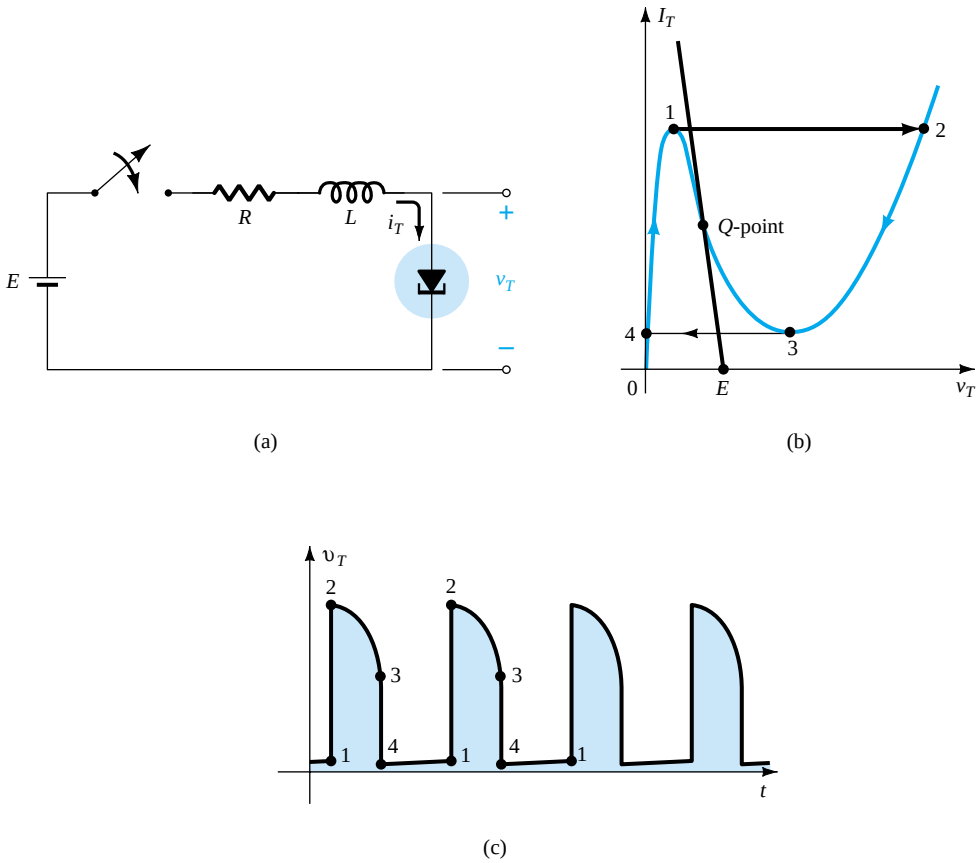


Figure 20.18 Negative-resistance oscillator.

$$E = I_T R + I_T(-R_T)$$

and

$$E = \underbrace{I_T}_{\text{less}}\underbrace{(R - R_T)}_{\text{less}}$$

If both elements of the equation above were to decrease, it would be impossible for the supply voltage to reach its set value. Therefore, for the current I_T to continue rising, the point of operation must shift from point 1 to point 2. However, at point 2, the voltage V_T has jumped to a value greater than the applied voltage (point 2 is to the right of any point on the network load line). To satisfy Kirchhoff's voltage law, the polarity of the transient voltage across the coil must reverse and the current begin to decrease as shown from 2 to 3 on the characteristics. When V_T drops to V_V, the characteristics suggest that the current I_T will begin to increase again. This is unacceptable since V_T is still more than the applied voltage and the coil is discharging through the series circuit. The point of operation must shift to point 4 to permit a continuation of the decrease in I_T. However, once at point 4, the potential levels are such that the tunnel current can again increase from 0 mA to I_P as shown on the characteristics. The process will repeat itself again and again, never settling in on the operating point defined for the unstable region. The resulting voltage across the tunnel diode appears in Fig. 20.18c and will continue as long as the dc supply is energized. The result is an oscillatory output established by a fixed supply and a device with a negative-resistance characteristic. The waveform of Fig. 20.18c has extensive application in timing and computer logic circuitry.

A tunnel diode can also be used to generate a sinusoidal voltage using simply a dc supply and a few passive elements. In Fig. 20.19a, the closing of the switch will result in a sinusoidal voltage that will decrease in amplitude with time. Depending on the elements employed, the time period can be from one almost instantaneous to one measurable in minutes using typical parameter values. This *damping* of the oscillatory output with time is due to the dissipative characteristics of the resistive elements. By placing a tunnel diode in series with the tank circuit as shown in Fig. 20.19c, the negative resistance of the tunnel diode will offset the resistive characteristics of the tank circuit, resulting in the *undamped* response appearing in the same figure. The

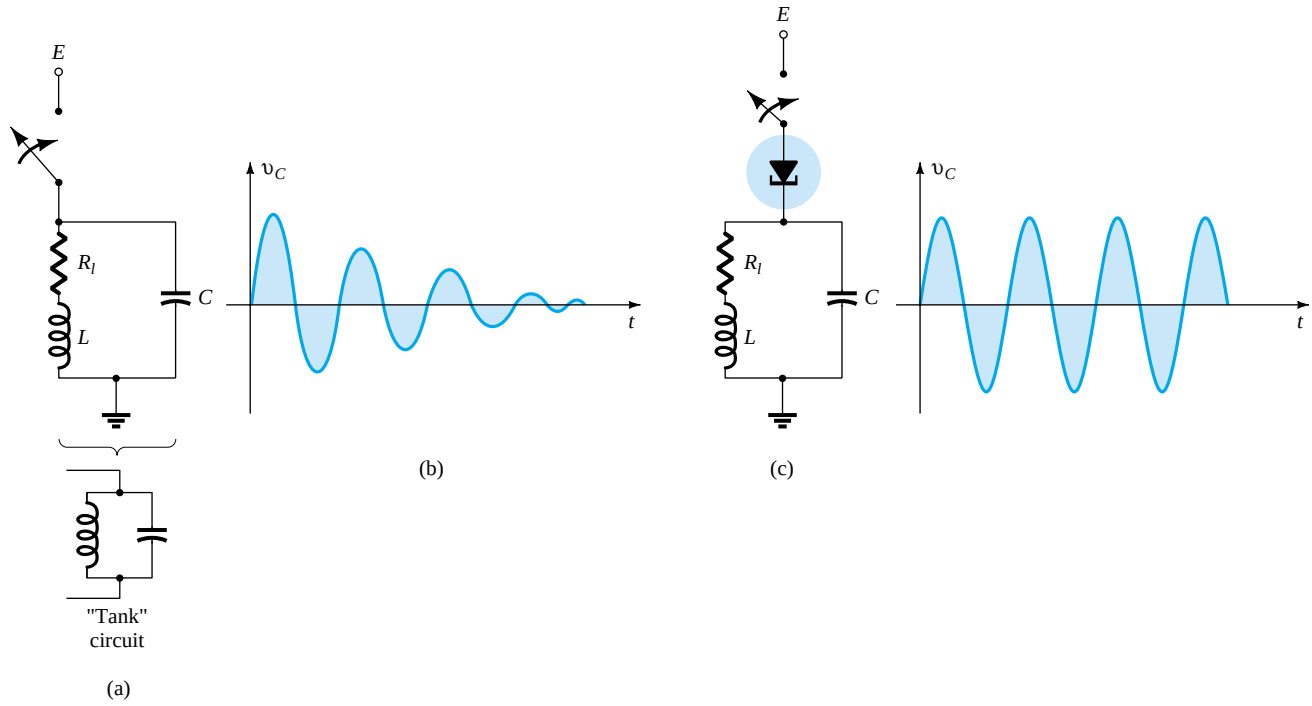


Figure 20.19 Sinusoidal oscillator.

design must continue to result in a load line that will intersect the characteristics only in the negative-resistance region. In another light, the sinusoidal generator of Fig. 20.19 is simply an extension of the pulse oscillator of Fig. 20.18, with the addition of the capacitor to permit an exchange of energy between the inductor and the capacitor during the various phases of the cycle depicted in Fig. 20.18b.

20.6 PHOTODIODES

The interest in light-sensitive devices has been increasing at an almost exponential rate in recent years. The resulting field of *optoelectronics* will be receiving a great deal of research interest as efforts are made to improve efficiency levels. Through the advertising media, the layperson has become quite aware that light sources offer a unique source of energy. This energy, transmitted as discrete packages called *photons*, has a level directly related to the frequency of the traveling light wave as determined by the following equation:

$$W = hf \quad \text{joules} \tag{20.5}$$

where h is called Planck's constant and is equal to 6.624×10^{-34} joule-second. It clearly states that since h is a constant, the energy associated with incident light waves is directly related to the frequency of the traveling wave.

The frequency is, in turn, directly related to the wavelength (distance between successive peaks) of the traveling wave by the following equation:

$$\lambda = \frac{v}{f} \tag{20.6}$$

where λ = wavelength, meters
v = velocity of light, 3×10^8 m/s
f = frequency of the traveling wave, hertz

The wavelength is usually measured in angstrom units (Å) or micrometers (μm), where

$$1 \text{ Å} = 10^{-10} \text{ m} \qquad \text{and} \qquad 1 \ \mu\text{m} = 10^{-6} \text{ m}$$

The wavelength is important because it will determine the material to be used in the optoelectronic device. The relative spectral response for Ge, Si, and selenium is provided in Fig. 20.20. The visible-light spectrum has also been included with an indication of the wavelength associated with the various colors.

The number of free electrons generated in each material is proportional to the *intensity* of the incident light. Light intensity is a measure of the amount of *luminous flux* falling in a particular surface area. Luminous flux is normally measured in *lumens* (lm) or watts. The two units are related by

$$1 \text{ lm} = 1.496 \times 10^{-10} \text{ W}$$

The light intensity is normally measured in lm/ft^2, footcandles (fc), or W/m^2, where

$$1 \text{ lm/ft}^2 = 1 \text{ fc} = 1.609 \times 10^{-9} \text{ W/m}^2$$

The photodiode is a semiconductor *p-n* junction device whose region of operation is limited to the reverse-bias region. The basic biasing arrangement, construction, and symbol for the device appear in Fig. 20.21.

Recall from Chapter 1 that the reverse saturation current is normally limited to a few microamperes. It is due solely to the thermally generated minority carriers in the

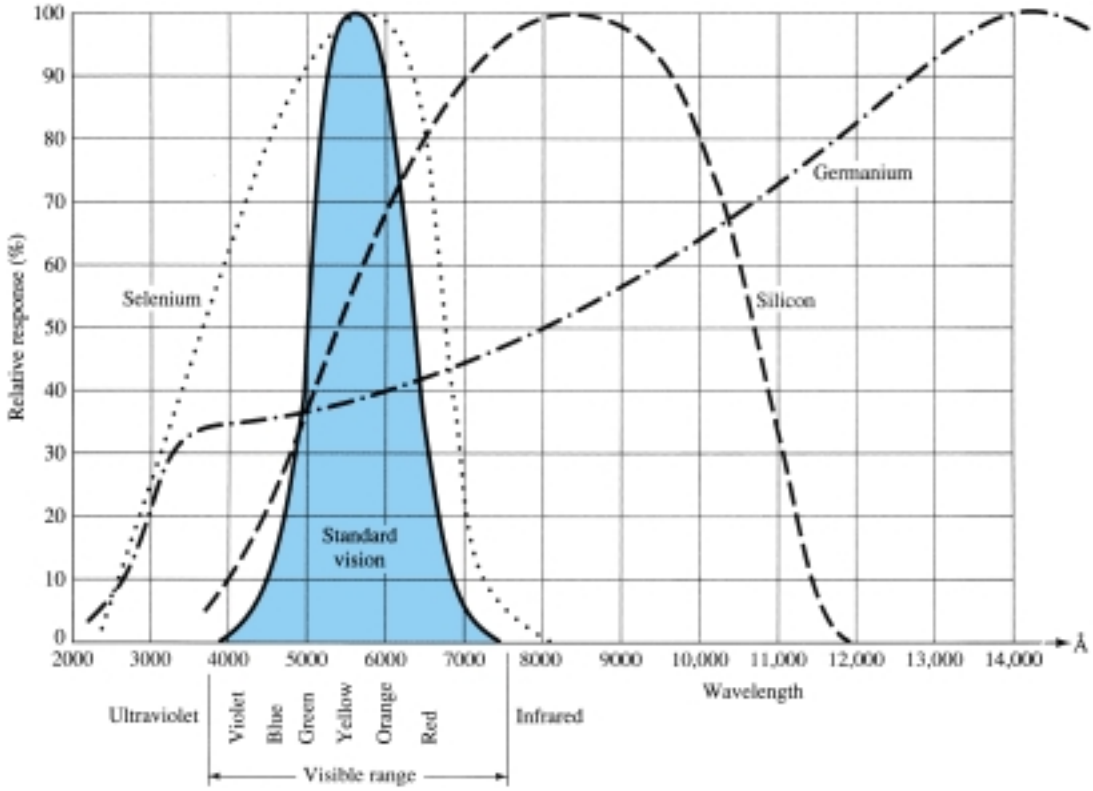


Figure 20.20 Relative spectral response for Si, Ge, and selenium as compared to the human eye.

n- and *p*-type materials. The application of light to the junction will result in a transfer of energy from the incident traveling light waves (in the form of photons) to the atomic structure, resulting in an increased number of minority carriers and an increased level of reverse current. This is clearly shown in Fig. 20.22 for different intensity levels. The *dark* current is that current that will exist with no applied illumination. Note that the current will only return to zero with a positive applied bias equal to V_T. In addition, Fig. 20.21 demonstrates the use of a lens to concentrate the light on the junction region. Commercially available photodiodes appear in Fig. 20.23.

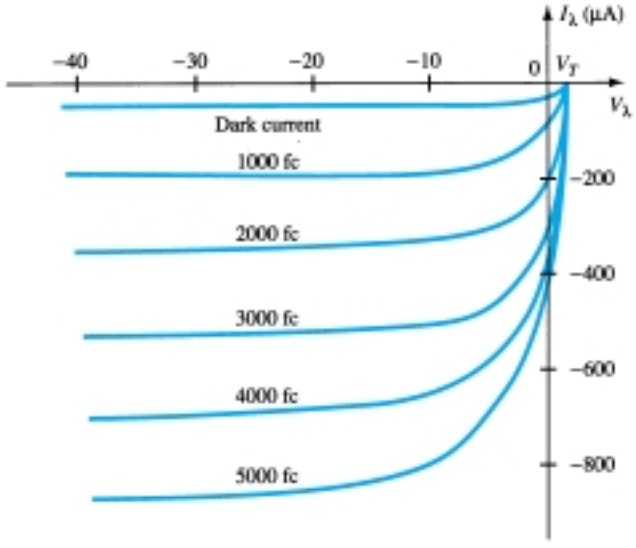


Figure 20.22 Photodiode characteristics.

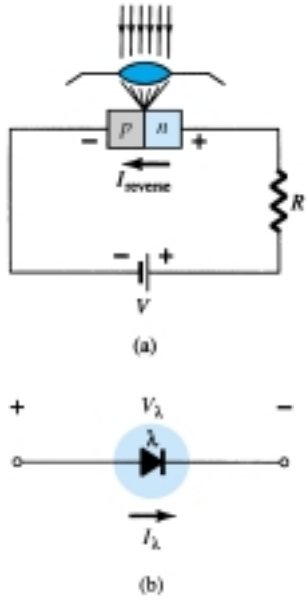


Figure 20.21 Photodiode: (a) basic biasing arrangement and construction; (b) symbol.

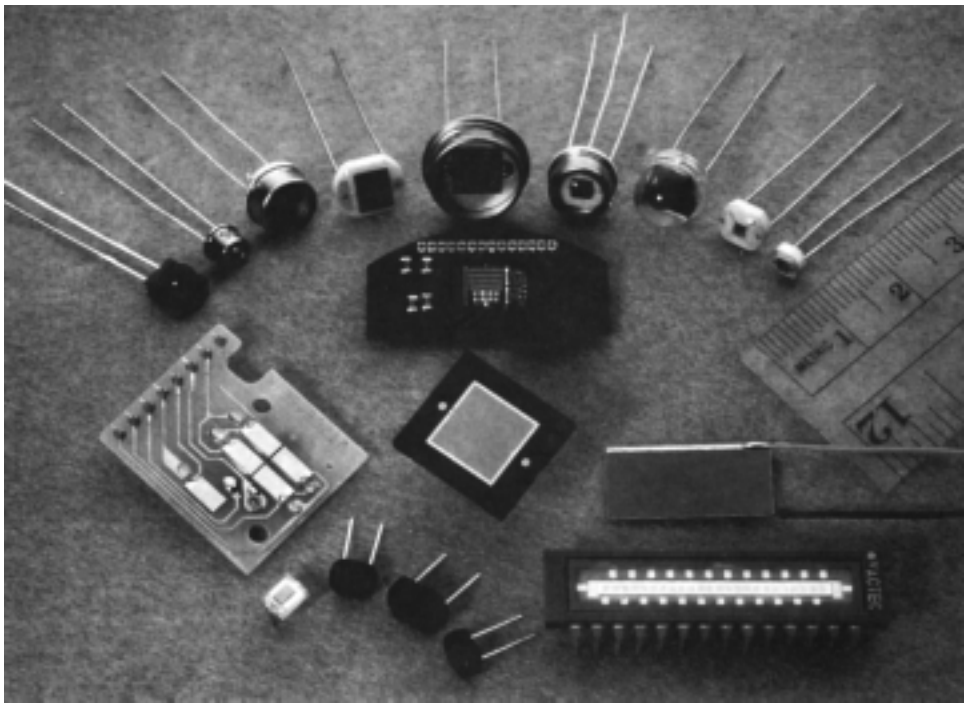


Figure 20.23 Photodiodes (Courtesy EG&G VACTEC, Inc.)

The almost equal spacing between the curves for the same increment in luminous flux reveals that the reverse current and luminous flux are almost linearly related. In other words, an increase in light intensity will result in a similar increase in reverse current. A plot of the two to show this linear relationship appears in Fig. 20.24 for a fixed voltage V_λ of 20 V. On the relative basis, we can assume that the reverse current is essentially zero in the absence of incident light. Since the rise and fall times (change-of-state parameters) are very small for this device (in the nanosecond range), the device can be used for high-speed counting or switching applications. Returning to Fig. 20.20, we note that Ge encompasses a wider spectrum of wavelengths than Si. This would make it suitable for incident light in the infrared region as provided by lasers and IR (infrared) light sources, to be described shortly. Of course, Ge has a higher dark current than silicon, but it also has a higher level of reverse current. The level of current generated by the incident light on a photodiode is not such that it could be used as a direct control, but it can be amplified for this purpose.

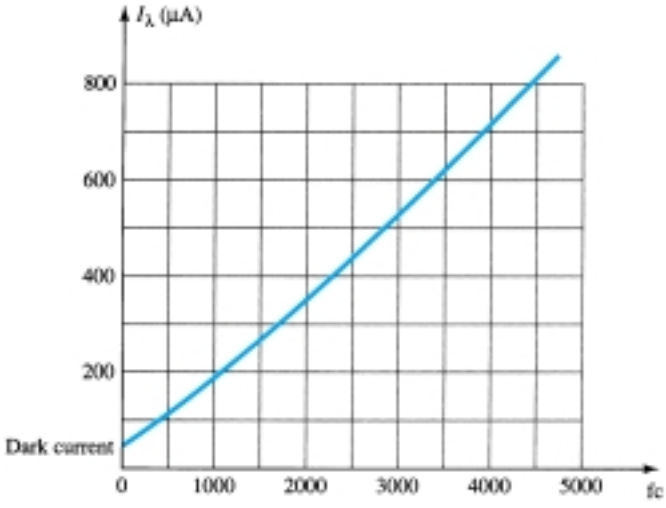


Figure 20.24 $I_\lambda(\mu A)$ versus fc (at V_λ = 20 V) for the photodiode of Fig. 20.22.

In Fig. 20.25, the photodiode is employed in an alarm system. The reverse current I_λ will continue to flow as long as the light beam is not broken. If interrupted, I_λ drops to the dark current level and sounds the alarm. In Fig. 20.26, a photodiode is used to count items on a conveyor belt. As each item passes the light beam is broken, I_λ drops to the dark current level and the counter is increased by one.

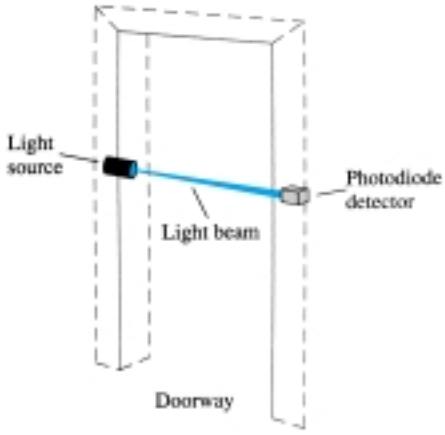


Figure 20.25 Using a photodiode in an alarm system.

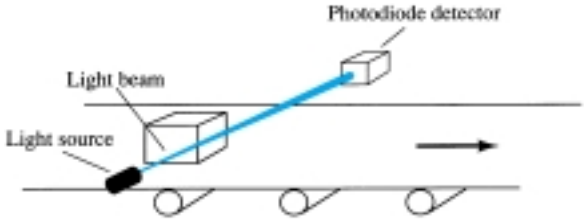
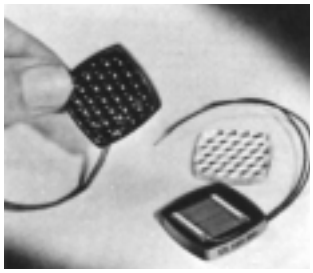


Figure 20.26 Using a photodiode in a counter operation.

20.7 PHOTOCONDUCTIVE CELLS

The photoconductive cell is a two-terminal semiconductor device whose terminal resistance will vary (linearly) with the intensity of the incident light. For obvious reasons, it is frequently called a *photoresistive device.* A typical photoconductive cell and the most widely used graphical symbol for the device appear in Fig. 20.27.



(a)



(b)

Figure 20.27 Photoconductive cell: (a) appearance; (b) symbol. [(a) Courtesy International Rectifier Corporation.]

The photoconductive materials most frequently used include cadmium sulfide (CdS) and cadmium selenide (CdSe). The peak spectral response of CdS occurs at approximately 5100 Å and for CdSe at 6150 Å (note Fig. 20.20). The response time of CdS units is about 100 ms, and 10 ms for CdSe cells. The photoconductive cell does not have a junction like the photodiode. A thin layer of the material connected between terminals is simply exposed to the incident light energy.

As the illumination on the device increases in intensity, the energy state of a larger number of electrons in the structure will also increase because of the increased availability of the photon packages of energy. The result is an increasing number of relatively "free" electrons in the structure and a decrease in the terminal resistance. The sensitivity curve for a typical photoconductive device appears in Fig. 20.28. Note the linearity (when plotted using a log-log scale) of the resulting curve and the large change in resistance (100 kΩ $\rightarrow$ 100 Ω) for the indicated change in illumination.

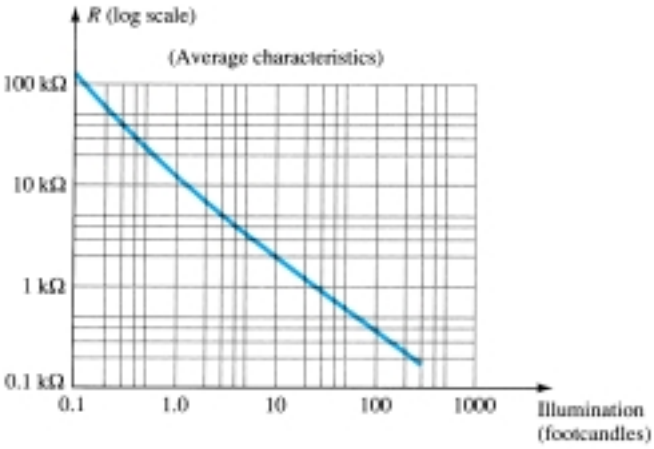


Figure 20.28 Photoconductive cell-terminal characteristics (GE type B425).

One rather simple, but interesting, application of the device appears in Fig. 20.29. The purpose of the system is to maintain V_o at a fixed level even though V_i may fluctuate from its rated value. As indicated in the figure, the photoconductive cell, bulb, and resistor all form part of this voltage-regulator system. If V_i should drop in magnitude for any number of reasons, the brightness of the bulb would also decrease. The decrease in illumination would result in an increase in the resistance (R_λ) of the photoconductive cell to maintain V_o at its rate level as determined by the voltage-divider rule, that is,

$$V_o = \frac{R_\lambda V_i}{R_\lambda + R_1} \tag{20.7}$$

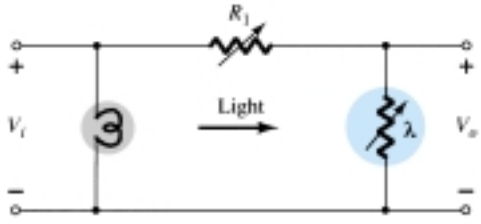
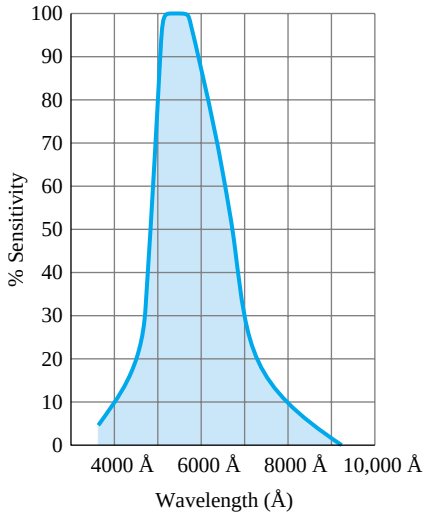


Figure 20.29 Voltage regulator employing a photoconductive cell.

In an effort to demonstrate the wealth of material available on each device from manufacturers, consider the CdS (cadmium sulfide) photoconductive cell described in Fig. 20.30. Note again the concern with temperature and response time.



Variation of Conductance With Temperature and Light

Footcandles	*0.01*	*0.1*	*1.0*	*10*	*100*
Temperature	*% Conductance*				
–25°C	103	104	104	102	106
0	98	102	102	100	103
25°C	100	100	100	100	100
50°C	98	102	103	104	99
75°C	90	106	108	109	104

Response Time Versus Light

Footcandles	*0.01*	*0.1*	*1.0*	*10*	*100*
Rise (seconds)	0.5	0.095	0.022	0.005	0.002
Decay (seconds)	0.125	0.021	0.005	0.002	0.001

Figure 20.30 Characteristics of a Clairex CdS photoconductive cell. (Courtesy Clairex Electronics.)

20.8 IR EMITTERS

Infrared-emitting diodes are solid-state gallium arsenide devices that emit a beam of radiant flux when forward-biased. The basic construction of the device is shown in Fig. 20.31. When the junction is forward-biased, electrons from the *n*-region will recombine with excess holes of the *p*-material in a specially designed recombination region sandwiched between the *p*- and *n*-type materials. During this recombination process, energy is radiated away from the device in the form of photons. The generated photons will either be reabsorbed in the structure or leave the surface of the device as radiant energy, as shown in Fig. 20.31.

The radiant flux in mW versus the dc forward current for a typical device appears in Fig. 20.32. Note the almost linear relationship between the two. An interesting pattern for such devices is provided in Fig. 20.33. Note the very narrow pattern for de-

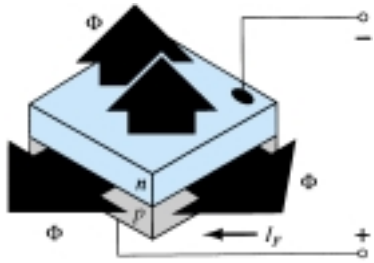


Figure 20.31 General structure of a semiconductor IR-emitting diode. (Courtesy RCA Solid State Division.)



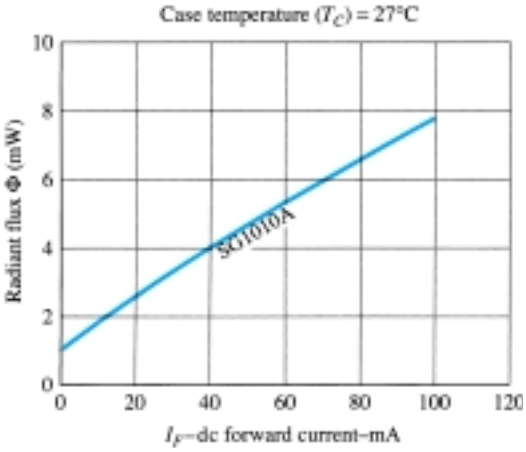


Figure 20.32 Typical radiant flux versus dc forward current for an IR-emitting diode. (Courtesy RCA Solid State Division.)

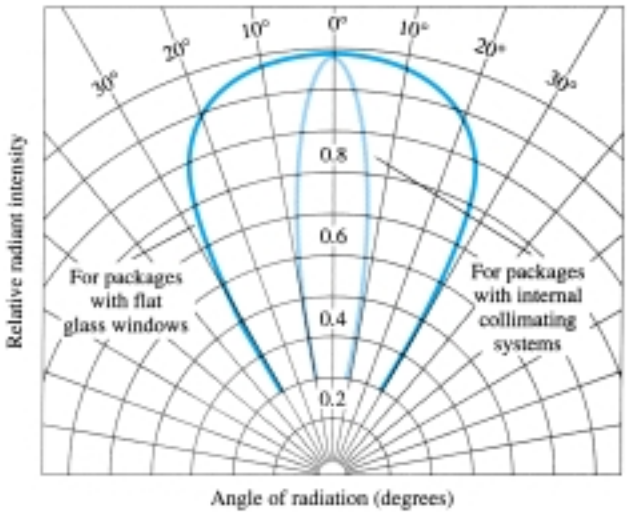
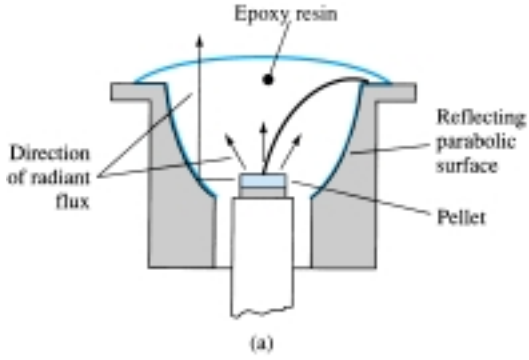


Figure 20.33 Typical radiant intensity patterns of RCA IR-emitting diodes. (Courtesy RCA Solid State Division.)

vices with an internal collimating system. One such device appears in Fig. 20.34, with its internal construction and graphical symbol. A few areas of application for such devices include card and paper-tape readers, shaft encoders, data-transmission systems, and intrusion alarms.





Approx. 2X actual size
(b)



Figure 20.34 RCA IR-emitting diode: (a) construction; (b) photo; (c) symbol. (Courtesy RCA Solid State Division.)

20.9 LIQUID-CRYSTAL DISPLAYS

The liquid-crystal display (LCD) has the distinct advantage of having a lower power requirement than the LED. It is typically in the order of microwatts for the display, as compared to the same order of milliwatts for LEDs. It does, however, require an external or internal light source and is limited to a temperature range of about 0° to 60°C. Lifetime is an area of concern because LCDs can chemically degrade. The types receiving the major interest today are the field-effect and dynamic-scattering units. Each will be covered in some detail in this section.

A liquid crystal is a material (normally organic for LCDs) that will flow like a liquid but whose molecular structure has some properties normally associated with solids. For the light-scattering units, the greatest interest is in the *nematic liquid crystal,* having the crystal structure shown in Fig. 20.35. The individual molecules have a rodlike appearance as shown in the figure. The indium oxide conducting surface is transparent, and under the condition shown in the figure, the incident light will simply pass through and the liquid-crystal structure will appear clear. If a voltage (for commercial units the threshold level is usually between 6 and 20 V) is applied across the conducting surfaces, as shown in Fig. 20.36, the molecular arrangement is disturbed, with the result that regions will be established with different indices of refraction. The incident light is therefore reflected in different directions at the interface between regions of different indices of refraction (referred to as *dynamic scattering*—first studied by RCA in 1968) with the result that the scattered light has a frosted-glass appearance. Note in Fig. 20.36, however, that the frosted look occurs

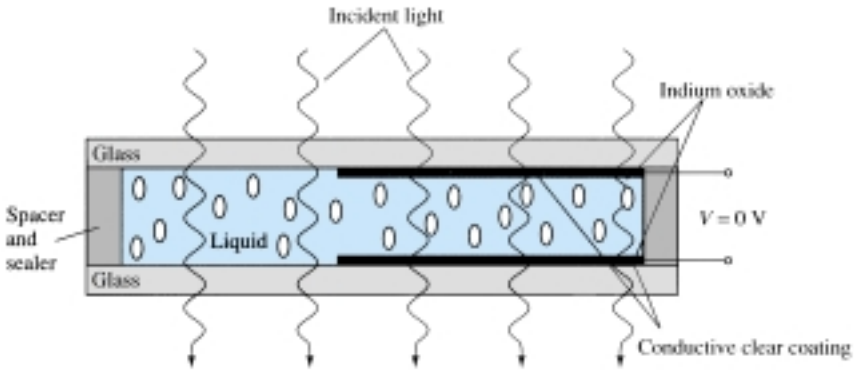


Figure 20.35 Nematic liquid crystal with no applied bias.

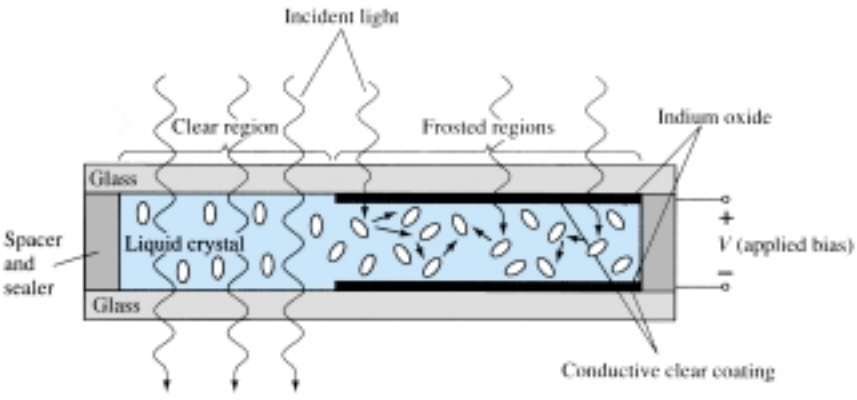


Figure 20.36 Nematic liquid crystal with applied bias.

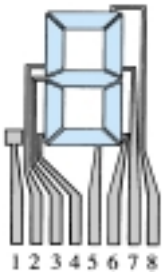


Figure 20.37 LCD eight-segment digit display.

only where the conducting surfaces are opposite each other and the remaining areas remain translucent.

A digit on an LCD display may have the segment appearance shown in Fig. 20.37. The black area is actually a clear conducting surface connected to the terminals below for external control. Two similar masks are placed on opposite sides of a sealed thick layer of liquid-crystal material. If the number 2 were required, the terminals 8, 7, 3, 4, and 5 would be energized, and only those regions would be frosted while the other areas would remain clear.

As indicated earlier, the LCD does not generate its own light but depends on an external or internal source. Under dark conditions, it would be necessary for the unit to have its own internal light source either behind or to the side of the LCD. During the day, or in lighted areas, a reflector can be put behind the LCD to reflect the light back through the display for maximum intensity. For optimum operation, current watch manufacturers are using a combination of the transmissive (own light source) and reflective modes called *transflective.*

The *field-effect* or *twisted nematic* LCD has the same segment appearance and thin layer of encapsulated liquid crystal, but its mode of operation is very different. Similar to the dynamic-scattering LCD, the field-effect LCD can be operated in the reflective or transmissive mode with an internal source. The transmissive display appears in Fig. 20.38. The internal light source is on the right, and the viewer is on the left. This figure is most noticeably different from Fig. 20.35 in that there is an addition of a *light polarizer.* Only the vertical component of the entering light on the right can pass through the vertical-light polarizer on the right. In the field-effect LCD, either the clear conducting surface to the right is chemically etched or an organic film is applied to orient the molecules in the liquid crystal in the vertical plane, parallel to the cell wall. Note the rods to the far right in the liquid crystal. The opposite conducting surface is also treated to ensure that the molecules are 90° out of phase in the direction shown (horizontal) but still parallel to the cell wall. In between the two walls of the liquid crystal there is a general drift from one polarization to the other, as shown in the figure. The left-hand light polarizer is also such that it permits the passage of only the vertically polarized incident light. If there is no applied voltage to the conducting surfaces, the vertically polarized light enters the liquid-crystal region and follows the 90° bending of the molecular structure. Its horizontal polarization at the left-hand vertical light polarizer does not allow it to pass through, and the viewer sees a uniformly dark pattern across the entire display. When a threshold voltage is applied (for commercial units from 2 to 8 V), the rodlike molecules align themselves with the field (perpendicular to the wall) and the light passes directly through without the 90° shift. The vertically incident light can then press directly through the second vertically polarized screen, and a light area is seen by the viewer. Through proper excita-

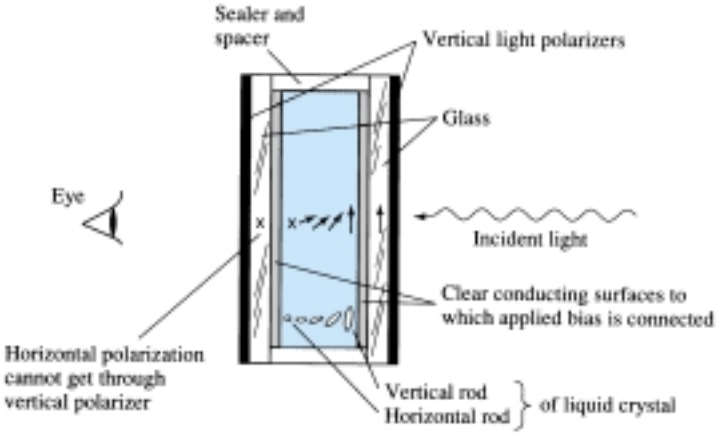


Figure 20.38 Transmissive field-effect LCD with no applied bias.

tion of the segments of each digit, the pattern will appear as shown in Fig. 20.39. The reflective-type field-effect LCD is shown in Fig. 20.40. In this case, the horizontally polarized light at the far left encounters a horizontally polarized filter and passes through to the reflector, where it is reflected back into the liquid crystal, bent back to the other vertical polarization, and returned to the observer. If there is no applied voltage, there is a uniformly lit display. The application of a voltage results in a vertically incident light encountering a horizontally polarized filter at the left, which it will not be able to pass through and will be reflected. A dark area results on the crystal, and the pattern as shown in Fig. 20.41 appears.

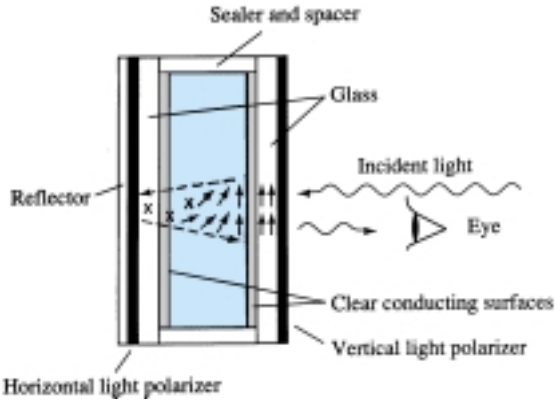


Figure 20.40 Reflective field-effect LCD with no applied bias.



Figure 20.39 Reflective-type LCD. (Courtesy RCA Solid State Division.)



Figure 20.41 Transmissive-type LCD. (Courtesy RCA Solid State Division.)

Field-effect LCDs are normally used when a source of energy is a prime factor (e.g., in watches, portable instrumentation, etc.) since they absorb considerably less power than the light-scattering types—the microwatt range compared to the low-milliwatt range. The cost is typically higher for field-effect units, and their height is limited to about 2 in. while light-scattering units are available up to 8 in. in height.

A further consideration in displays is turn-on and turn-off time. LCDs are characteristically much slower then LEDs. LCDs typically have response times in the range 100 to 300 ms, while LEDs are available with response times below 100 ns. However, there are numerous applications, such as in a watch, where the difference between 100 ns and 100 ms ($\frac{1}{10}$ of a second) is of little consequence. For such applications, the lower power demand of LCDs is a very attractive characteristic. The lifetime of LCD units is steadily increasing beyond the 10,000+ hours limit. Since the color generated by LCD units is dependent on the source of illumination, there is a greater range of color choice.

20.10 SOLAR CELLS

In recent years, there has been increasing interest in the solar cell as an alternative source of energy. When we consider that the power density received from the sun at sea level is about 100 mW/cm^2 (1 kW/m^2), it is certainly an energy source that requires further research and development to maximize the conversion efficiency from solar to electrical energy.

The basic construction of a silicon *p-n* junction solar cell appears in Fig. 20.42. As shown in the top view, every effort is made to ensure that the surface area perpendicular to the sun is a maximum. Also, note that the metallic conductor connected to the *p*-type material and the thickness of the *p*-type material are such that they ensure that a maximum number of photons of light energy will reach the junction. A photon of light energy in this region may collide with a valence electron and impart to it sufficient energy to leave the parent atom. The result is a generation of free electrons and holes. This phenomenon will occur on each side of the junction. In the



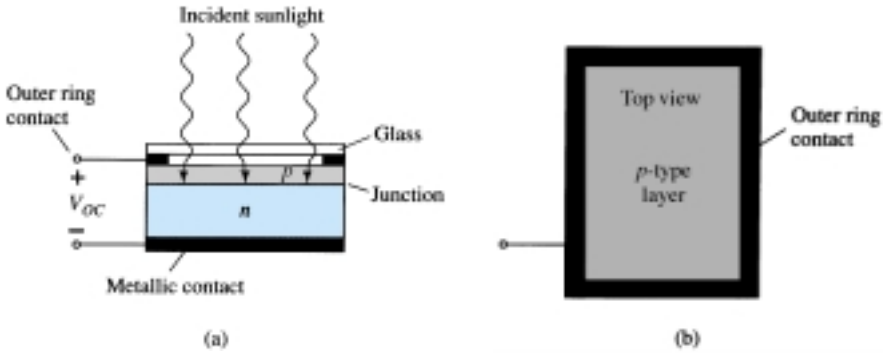


Figure 20.42 Solar cell: (a) cross section; (b) top view.

p-type material, the newly generated electrons are minority carriers and will move rather freely across the junction as explained for the basic *p-n* junction with no applied bias. A similar discussion is true for the holes generated in the *n*-type material. The result is an increase in the minority-carrier flow, which is opposite in direction to the conventional forward current of a *p-n* junction. This increase in reverse current is shown in Fig. 20.43. Since $V = 0$ anywhere on the vertical axis and represents a short-circuit condition, the current at this intersection is called the *short-circuit current* and is represented by the notation I_{SC}. Under open-circuit conditions ($i_d = 0$), the *photovoltaic* voltage V_{OC} will result. This is a logarithmic function of the illumination, as shown in Fig. 20.44. V_{OC} is the terminal voltage of a battery under no-load (open-circuit) conditions. Note, however, in the same figure that the short-circuit current is a linear function of the illumination. That is, it will double for the same increase in illumination (f_{C_1} and $2f_{C_1}$ in Fig. 20.44) while the change in V_{OC} is less for this region. The major increase in V_{OC} occurs for lower-level increases in illumination. Eventually, a further increase in illumination will have very little effect on V_{OC}, although I_{SC} will increase, causing the power capabilities to increase.

Selenium and silicon are the most widely used materials for solar cells, although gallium arsenide, indium arsenide, and cadmium sulfide, among others, are also used.

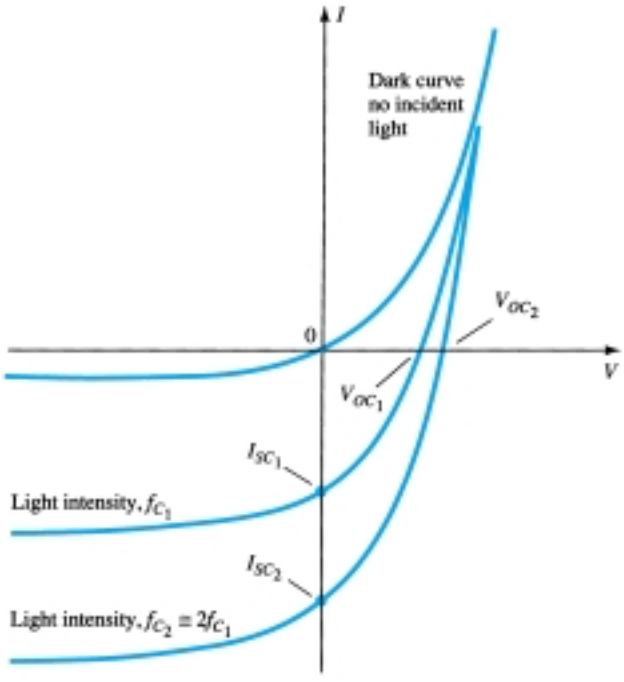


Figure 20.43 Short-circuit current and open-circuit voltage versus light intensity for a solar cell.

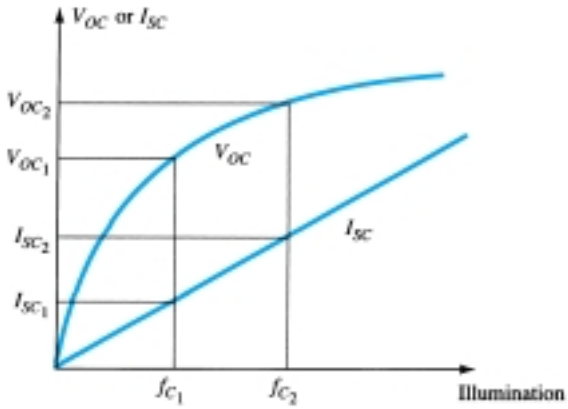


Figure 20.44 V_{OC} and I_{SC} versus illumination for a solar cell.

The wavelength of the incident light will affect the response of the *p-n* junction to the incident photons. Note in Fig. 20.45 how closely the selenium cell response curve matches that of the eye. This fact has widespread application in photographic equipment such as exposure meters and automatic exposure diaphragms. Silicon also overlaps the visible spectrum but has its peak at the 0.8 μm (8000 Å) wavelength, which is in the infrared region. In general, silicon has a higher conversion efficiency, greater stability, and is less subject to fatigue. Both materials have excellent temperature characteristics. That is, they can withstand extreme high or low temperatures without a significant drop-off in efficiency. Typical solar cells, with their electrical characteristics, appear in Fig. 20.46.

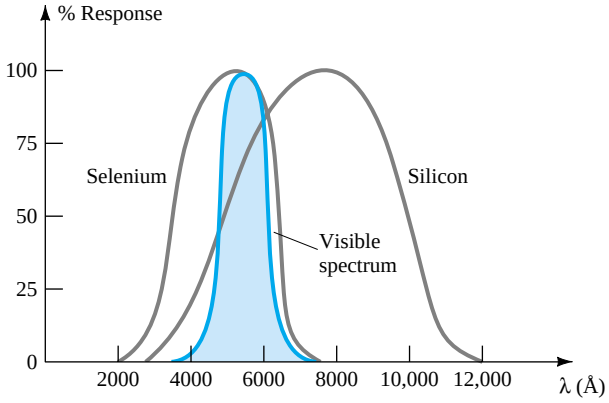


Figure 20.45 Spectral response of Se, Si, and the naked eye.

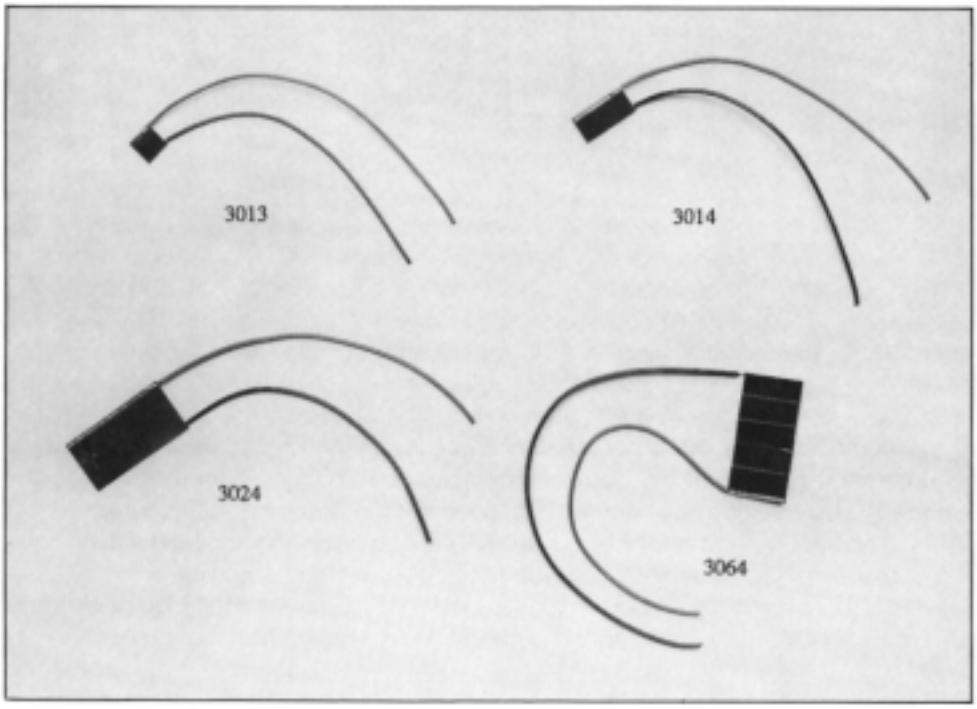
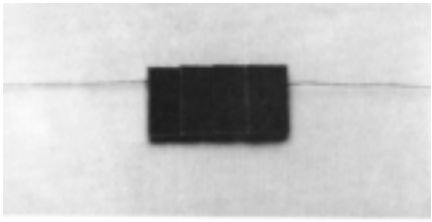


Figure 20.46 Typical solar cells and their electrical characteristics. (Courtesy EG&G VACTEC, Inc.)

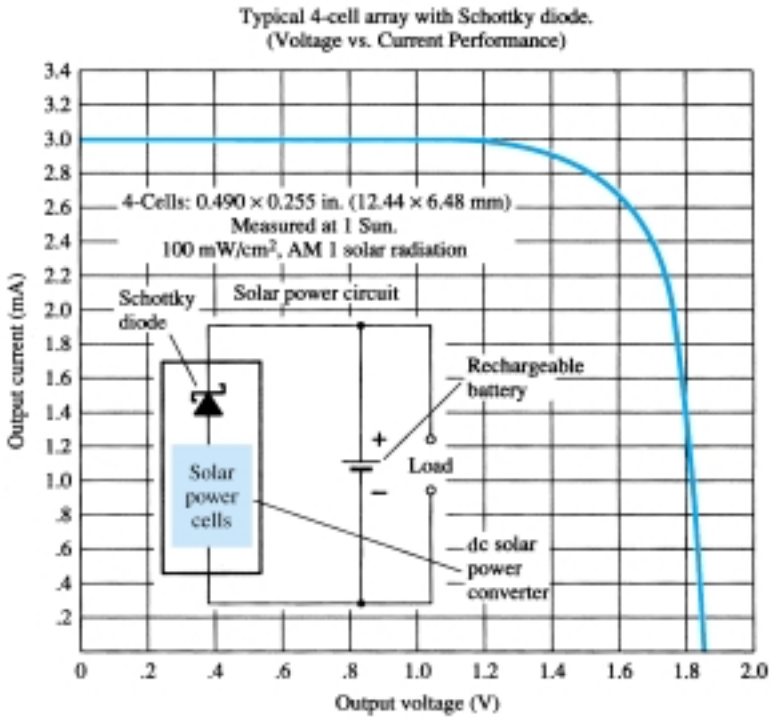
Electrical Characteristics

Part No.	Active Area	Test Voltage	Minimum current Test voltage
3013	0.032 in² (0.21 cm²)	0.4 V	4.2 mA
3014	0.065 in² (0.42 cm²)	0.4 V	8.4 mA
3024	0.29 in² (1.87 cm²)	0.4 V	38 mA
3064	0.325 in² (2.1 cm²)	2 V	8.4 mA

A very recent innovation in the use of solar cells appears in Fig. 20.47. The series arrangement of solar cells permits a voltage beyond that of a single element. The performance of a typical four-cell array appears in the same figure. At a current of approximately 2.6 mA, the output voltage is about 1.6 V, resulting in an output power of 4.16 mW. The Schottky barrier diode is included to prevent battery current drain through the power converter. That is, the resistance of the Schottky diode is so high to charge flowing down through (+ to −) the power converter that it will appear as an open circuit to the rechargeable battery and not draw current from it.



(a)



(b)

Figure 20.47 International Rectifier four-cell array: (a) appearance; (b) characteristics. (Courtesy International Rectifier Corporation.)

It might be of interest to note that the Lockheed Missiles and Space Company is working on a grant from the National Aeronautics and Space Administration to develop a massive solar-array wing for the space shuttle. The wing will measure 13.5 ft by 105 ft when extended and will contain 41 panels, each carrying 3060 silicon solar cells. The wing can generate a total of 12.5 kW of electrical power.

The efficiency of operation of a solar cell is determined by the electrical power output divided by the power provided by the light source. That is,

$$\eta = \frac{P_{o(\text{electrical})}}{P_{i(\text{light energy})}} \times 100\% = \frac{P_{\max(\text{device})}}{(\text{area in cm}^2)(100\ \text{mW/cm}^2)} \times 100\% \qquad (20.8)$$

Typical levels of efficiency range from 10% to 40%—a level that should improve measurably if the present interest continues. A typical set of output characteristics for silicon solar cells of 10% efficiency with an active area of 1 cm^2 appears in Fig. 20.48. Note the optimum power locus and the almost linear increase in output current with luminous flux for a fixed voltage.

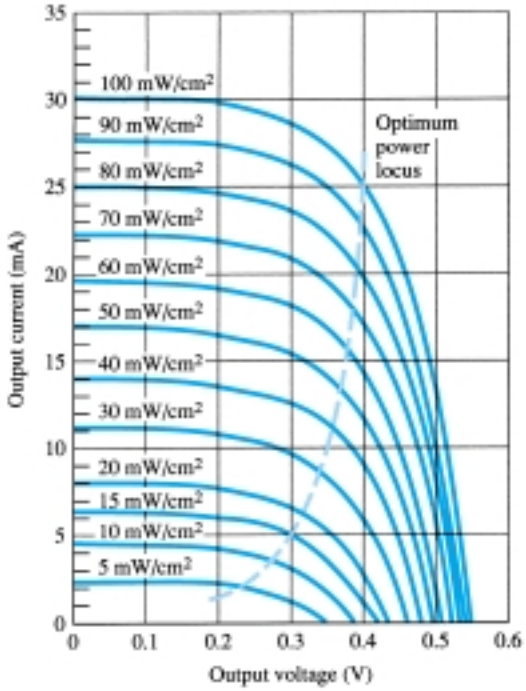


Figure 20.48 Typical output characteristics for silicon solar cells of 10% efficiency having an active area of 1 cm^2. Cell temperature is 30°C.

20.11 THERMISTORS

The thermistor is, as the name implies, a temperature-sensitive resistor; that is, its terminal resistance is related to its body temperature. It is not a junction device and is constructed of Ge, Si, or a mixture of oxides of cobalt, nickel, strontium, or manganese. The compound employed will determine whether the device has a positive or negative temperature coefficient.

The characteristics of a representative thermistor with a negative temperature coefficient are provided in Fig. 20.49, with the commonly used symbol for the device. Note in particular that at room temperature (20°C) the resistance of the thermistor is approximately 5000 Ω, while at 100°C (212°F) the resistance has decreased to 100 Ω. A temperature span of 80°C has therefore resulted in a 50:1 change in resis-

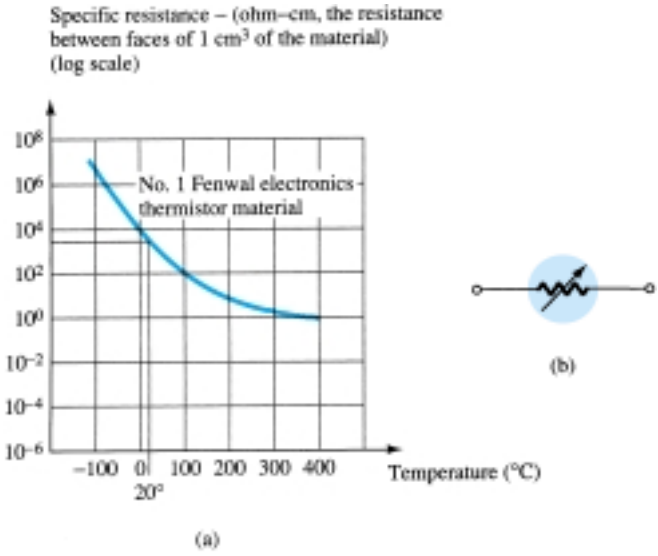


Figure 20.49 Thermistor: (a) typical set of characteristics; (b) symbol.

tance. It is typically 3% to 5% per degree change in temperature. There are fundamentally two ways to change the temperature of the device: internally and externally. A simple change in current through the device will result in an internal change in temperature. A small applied voltage will result in a current too small to raise the body temperature above that of the surroundings. In this region, as shown in Fig. 20.50, the thermistor will act like a resistor and have a positive temperature coefficient. However, as the current increases, the temperature will rise to the point where the negative temperature coefficient will appear as shown in Fig. 20.50. The fact that the rate of internal flow can have such an effect on the resistance of the device introduces a wide vista of applications in control, measuring techniques, and so on. An external change would require changing the temperature of the surrounding medium or immersing the device in a hot or cold solution.

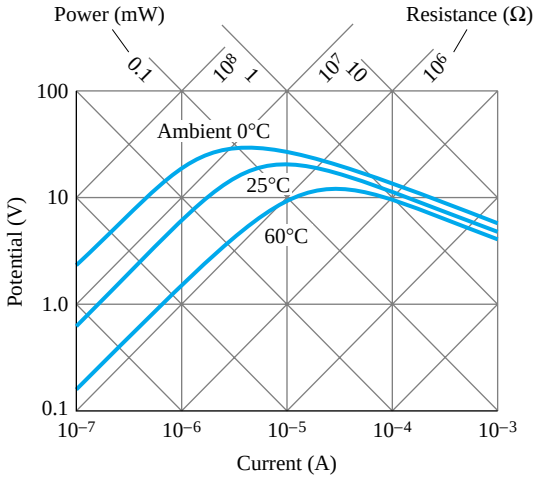


Figure 20.50 Steady-state voltage–current characteristics of Fenwal Electronics BK65VI Thermistor. (Courtesy Fenwal Electronics, Incorporated.)

A photograph of a number of commercially available thermistors is provided in Fig. 20.51. A simple temperature-indicating circuit appears in Fig. 20.52. Any increase in the temperature of the surrounding medium will result in a decrease in the resistance of the thermistor and an increase in the current I_T. An increase in I_T will

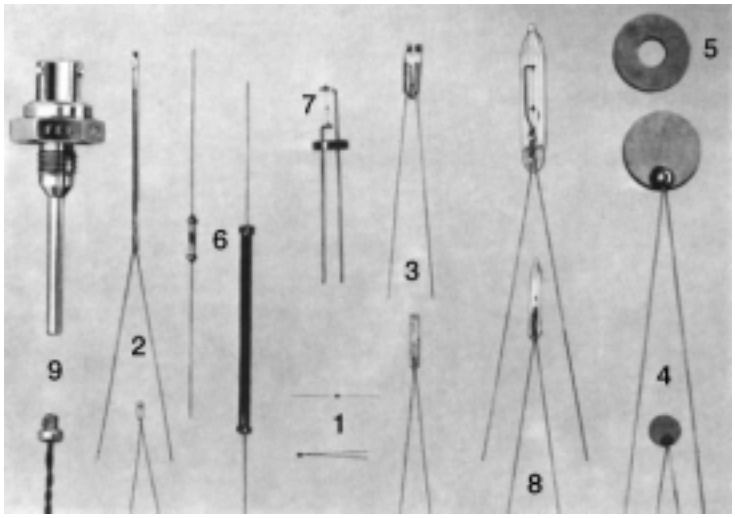


Figure 20.51 Various types of thermistors: (1) beads; (2) glass probes; (3) iso-curve interchangeable probes and beads; (4) disks; (5) washers; (6) rods; (7) specially mounted beads; (8) vacuum and gas-filled probes; (9) special probe assemblies. (Courtesy Fenwal Electronics, Incorporated.)

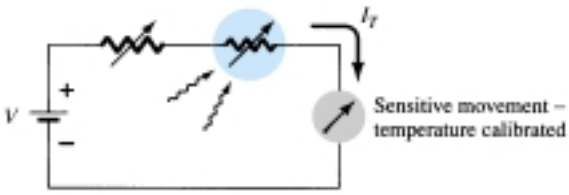


Figure 20.52 Temperature-indicating circuit.

produce an increased movement deflection, which when properly calibrated will accurately indicate the higher temperature. The variable resistance was added for calibration purposes.

PROBLEMS

§ 20.2 Schottky Barrier (Hot-Carrier) Diodes

1. (a) Describe in your own words how the construction of the hot-carrier diode is significantly different from the conventional semiconductor diode.
(b) In addition, describe its mode of operation.

2. (a) Consult Fig. 20.2. How would you compare the dynamic resistances of the diodes in the forward-bias regions?
(b) How do the levels of I_s and V_Z compare?

3. Referring to Fig. 20.5, how does the maximum surge current I_{FSM} relate to the average rectified forward current? Is it typically greater than 20:1? Why is it possible to have such high levels of current? What noticeable difference is there in construction as the current rating increases?

4. Referring to Fig. 20.6a, at what temperature is the forward voltage drop 300 mV at a current of 1 mA? Which current levels have the highest levels of temperature coefficients? Assume a linear progression between temperature levels.

* **5.** For the curve of Fig. 20.6b denoted 2900/2303, determine the percent change in I_R for a change in reverse voltage from 5 to 10 V. At what reverse voltage would you expect to reach a reverse current of 1 μA? Note the log scale for I_R.

* **6.** Determine the percent change in capacitance between 0 and 2 V for the 2900/2303 curve of Fig. 20.6c. How does this compare to the change between 8 and 10 V?

§ 20.3 Varactor (Varicap) Diodes

7. (a) Determine the transition capacitance of a diffused junction varicap diode at a reverse potential of 4.2 V if $C(0) = 80$ pF and $V_T = 0.7$ V.
(b) From the information of part (a), determine the constant K in Eq. (20.2).

8. (a) For a varicap diode having the characteristics of Fig. 20.7, determine the difference in capacitance between reverse-bias potentials of -3 and -12 V.
(b) Determine the incremental rate of change ($\Delta C/\Delta V_r$) at $V = -8$ V. How does this value compare with the incremental change determined at -2 V?

* **9.** (a) The resonant frequency of a series RLC network is determined by $f_0 = 1/(2\pi\sqrt{LC})$. Using the value of f_0 and L_S provided in Fig. 20.9, determine the value of C.
(b) How does the value calculated in part (a) compare with that determined by the curve in Fig. 20.10 at $V_R = 25$ V?

10. Referring to Fig. 20.10, determine the ratio of capacitance at $V_R = 3$ V to $V_R = 25$ V and compare to the value of C_3/C_{25} given in Fig. 20.9 (maximum = 6.5).

11. Determine T_1 for a varactor diode if $C_0 = 22$ pF, $TC_C = 0.02\%/°C$, and $\Delta C = 0.11$ pF due to an increase in temperature above $T_0 = 25°C$.

12. What region of V_R would appear to have the greatest change in capacitance per change in reverse voltage for the BB139 varactor diode of Figs. 20.9 and 20.10? Be aware that the scales are nonlinear.

* **13.** If $Q = X_L/R = 2\pi fL/R$, determine the figure of merit (Q) at 600 MHz using the fact that $R_S = 0.35\ \Omega$ and $L_S = 2.5$ nH. Comment on the change in Q with frequency and the support or nonsupport of the curve in Fig. 20.10.

§ 20.4 Power Diodes

14. Consult a manufacturer's data book and compare the general characteristics of a high-power device (>10 A) to a low-power unit (<100 mA). Is there a significant change in the data and characteristics provided? Why?

§ 20.5 Tunnel Diodes

15. What are the essential differences between a semiconductor junction diode and a tunnel diode?

***16.** Note in the equivalent circuit of Fig. 20.14 that the capacitor appears in parallel with the negative resistance. Determine the reactance of the capacitor at 1 MHz and 100 MHz if $C = 5$ pF, and determine the total impedance of the parallel combination (with $R = -152\ \Omega$) at each frequency. Is the magnitude of the inductive reactance anything to be overly concerned about at either of these frequencies if $L_S = 6$ nH?

***17.** Why do you believe the maximum reverse current rating for the tunnel diode can be greater than the forward current rating? (*Hint:* Note the characteristics and consider the power rating.)

18. Determine the negative resistance for the tunnel diode of Fig. 20.13 between $V_T = 0.1$ V and $V_T = 0.3$ V.

19. Determine the stable operating points for the network of Fig. 20.17 if $E = 2$ V, $R = 0.39$ kΩ, and the tunnel diode of Fig. 20.13 is employed. Use typical values from Table 20.1.

***20.** For $E = 0.5$ V and $R = 51\ \Omega$, sketch v_T for the network of Figure 20.18 and the tunnel diode of Fig. 20.13.

21. Determine the frequency of oscillation for the network of Fig. 20.19 if $L = 5$ mH, $R_l = 10\ \Omega$, and $C = 1\ \mu$F.

§ 20.6 Photodiodes

22. Determine the energy associated with the photons of green light if the wavelength is 5000 Å. Give your answer in joules and electron volts.

23. (a) Referring to Fig. 20.20, what would appear to be the frequencies associated with the upper and lower limits of the visible spectrum?
(b) What is the wavelength in microns associated with the peak relative response of silicon?
(c) If we define the bandwidth of the spectral response of each material to occur at 70% of its peak level, what is the bandwidth of silicon?

24. Referring to Fig. 20.22, determine I_λ if $V_\lambda = 30$ V and the light intensity is 4×10^{-9} W/m^2.

25. (a) Which material of Fig. 20.20 would appear to provide the best response to yellow, red, green, and infrared (less than 11,000 Å) light sources?
(b) At a frequency of 0.5×10^{15} Hz, which color has the maximum spectral response?

***26.** Determine the voltage drop across the resistor of Fig. 20.21 if the incident flux is 3000 fc, $V_\lambda = 25$ V, and $R = 100$ kΩ. Use the characteristics of Fig. 20.22.

§ 20.7 Photoconductive Cells

***27.** What is the approximate rate of change of resistance with illumination for a photoconductive cell with the characteristics of Fig. 20.28 for the ranges (a) $0.1 \rightarrow 1$ kΩ, (b) $1 \rightarrow 10$ kΩ, and (c) $10 \rightarrow$ kΩ? (Note that this is a log scale.) Which region has the greatest rate of change in resistance with illumination?

28. What is the "dark current" of a photodiode?

29. If the illumination on the photoconductive diode in Fig. 20.29 is 10 fc, determine the magnitude of V_i to establish 6 V across the cell if R_1 is equal to 5 kΩ. Use the characteristics of Fig. 20.28.

***30.** Using the data provided in Fig. 20.30, sketch a curve of percent conductance versus temperature for 0.01, 1.0, and 100 fc. Are there any noticeable effects?

***31.** (a) Sketch a curve of rise time versus illumination using the data from Fig. 20.30.
(b) Repeat part (a) for the decay time.
(c) Discuss any noticeable effects of illumination in parts (a) and (b).

32. Which colors is the CdS unit of Fig. 20.30 most sensitive to?

§ 20.8 IR Emitters

33. (a) Determine the radiant flux at a dc forward current of 70 mA for the device of Fig. 20.32.
(b) Determine the radiant flux in lumens at a dc forward current of 45 mA.

***34.** (a) Through the use of Fig. 20.33, determine the relative radiant intensity at an angle of 25° for a package with a flat glass window.
(b) Plot a curve of relative radiant intensity versus degrees for the flat package.

***35.** If 60 mA of dc forward current is applied to an SG1010A IR emitter, what will be the incident radiant flux in lumens 5° off the center if the package has an internal collimating system? Refer to Figs. 20.32 and 20.33.

§ 20.9 Liquid-Crystal Displays

36. Referring to Fig. 20.37, which terminals must be energized to display number 7?

37. In your own words, describe the basic operation of an LCD.

38. Discuss the relative differences in mode of operation between an LED and an LCD display.

39. What are the relative advantages and disadvantages of an LCD display as compared to an LED display?

§ 20.10 Solar Cells

40. A 1-cm by 2-cm solar cell has a conversion efficiency of 9%. Determine the maximum power rating of the device.

***41.** If the power rating of a solar cell is determined on a very rough scale by the product $V_{OC}\, I_{SC}$, is the greatest rate of increase obtained at lower or higher levels of illumination? Explain your reasoning.

42. (a) Referring to Fig. 20.48, what power density is required to establish a current of 24 mA at an output voltage of 0.25 V?
(b) Why is 100 mW/cm^2 the maximum power density in Fig. 20.48?
(c) Determine the output current if the power is 40 mW/cm^2 and the output voltage is 0.3 V.

***43.** (a) Sketch a curve of output current versus power density at an output voltage of 0.15 V using the characteristics of Fig. 20.48.
(b) Sketch a curve of output voltage versus power density at a current of 19 mA.
(c) Is either of the curves from parts (a) and (b) linear within the limits of the maximum power limitation?

§ 20.11 Thermistors

***44.** For the thermistor of Fig. 20.49, determine the dynamic rate of change in specific resistance with temperature at $T = 20°C$. How does this compare to the value determined at $T = 300°C$? From the results, determine whether the greatest change in resistance per unit change in temperature occurs at lower or higher levels of temperature. Note the vertical log scale.

45. Using the information provided in Fig. 20.49, determine the total resistance of a 2-cm length of the material having a perpendicular surface area of 1 cm^2 at a temperature of 0°C. Note the vertical log scale.

46. (a) Referring to Fig. 20.50, determine the current at which a 25°C sample of the material changes from a positive to negative temperature coefficient. (Figure 20.50 is a log scale.)
(b) Determine the power and resistance levels of the device (Fig. 20.50) at the peak of the 0°C curve.
(c) At a temperature of 25°C, determine the power rating if the resistance level is 1 MΩ.

47. In Fig. 20.52, $V = 0.2$ V and $R_{\text{variable}} = 10\ \Omega$. If the current through the sensitive movement is 2 mA and the voltage drop across the movement is 0 V, what is the resistance of the thermistor?

*Please Note: Asterisks indicate more difficult problems.